GATES LEARJET CORPORATION

maintenance manual

CHAPTER

25

EQUIPMENT / FURNISHINGS

RECORD OF TEMPORARY REVISIONS

Rev. No.	Date Inserted	Date Removed	Page Number	Rev. No.	Date Inserted	Date Removed	Page Number
25-1	May 3/78	Rev #14 Apr 30/82	25-30-01 Page 201				
25-2	Oct 31/80	Rev #14 Apr 30/82	25-62-00 Page 201				

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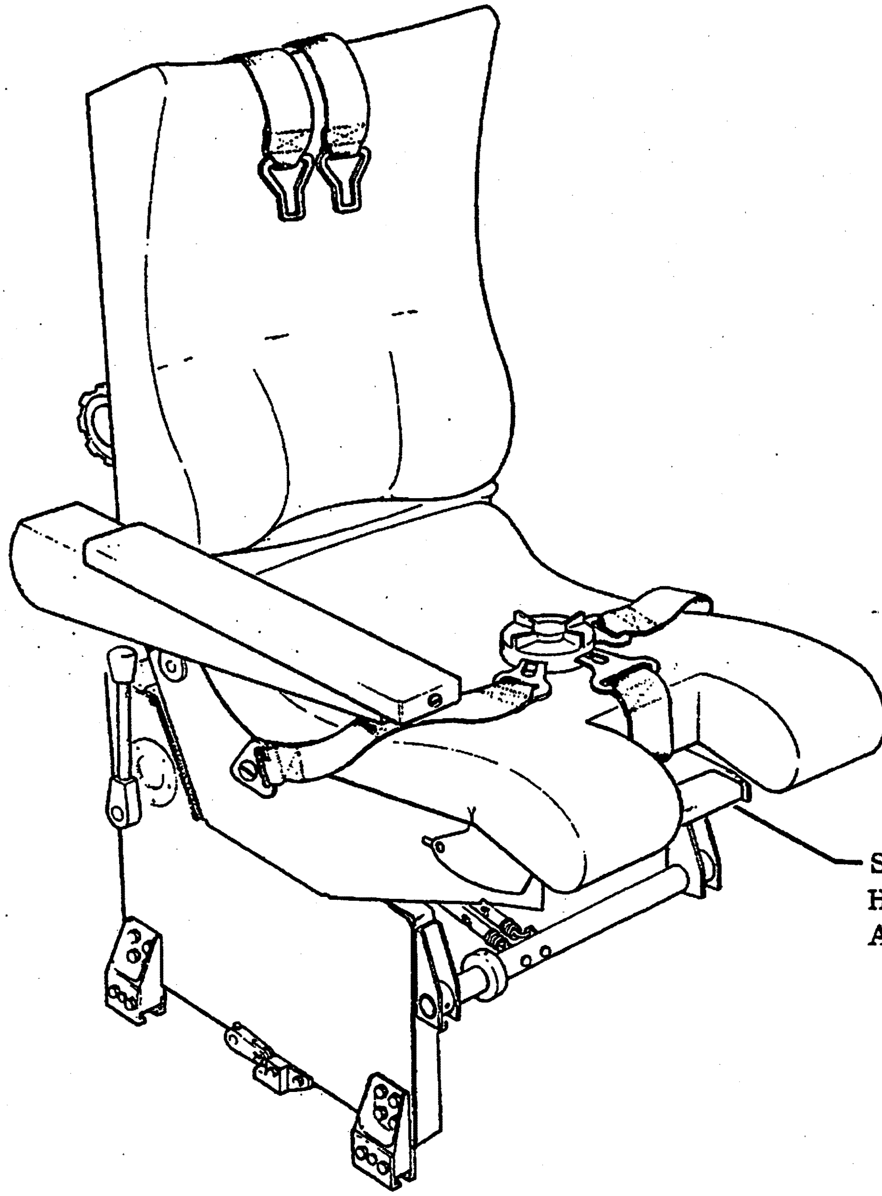
FLIGHT COMPARTMENT - DESCRIPTION AND OPERATION

1. General
   A. The flight compartment is divided from the passenger compartment by the service cabinets directly behind the pilot's and copilot's seats.
   B. Equipment and furnishings within the flight compartment include seats, seat belts, shoulder harnesses, emergency hatchet, emergency tool kit, life vests and a map case.

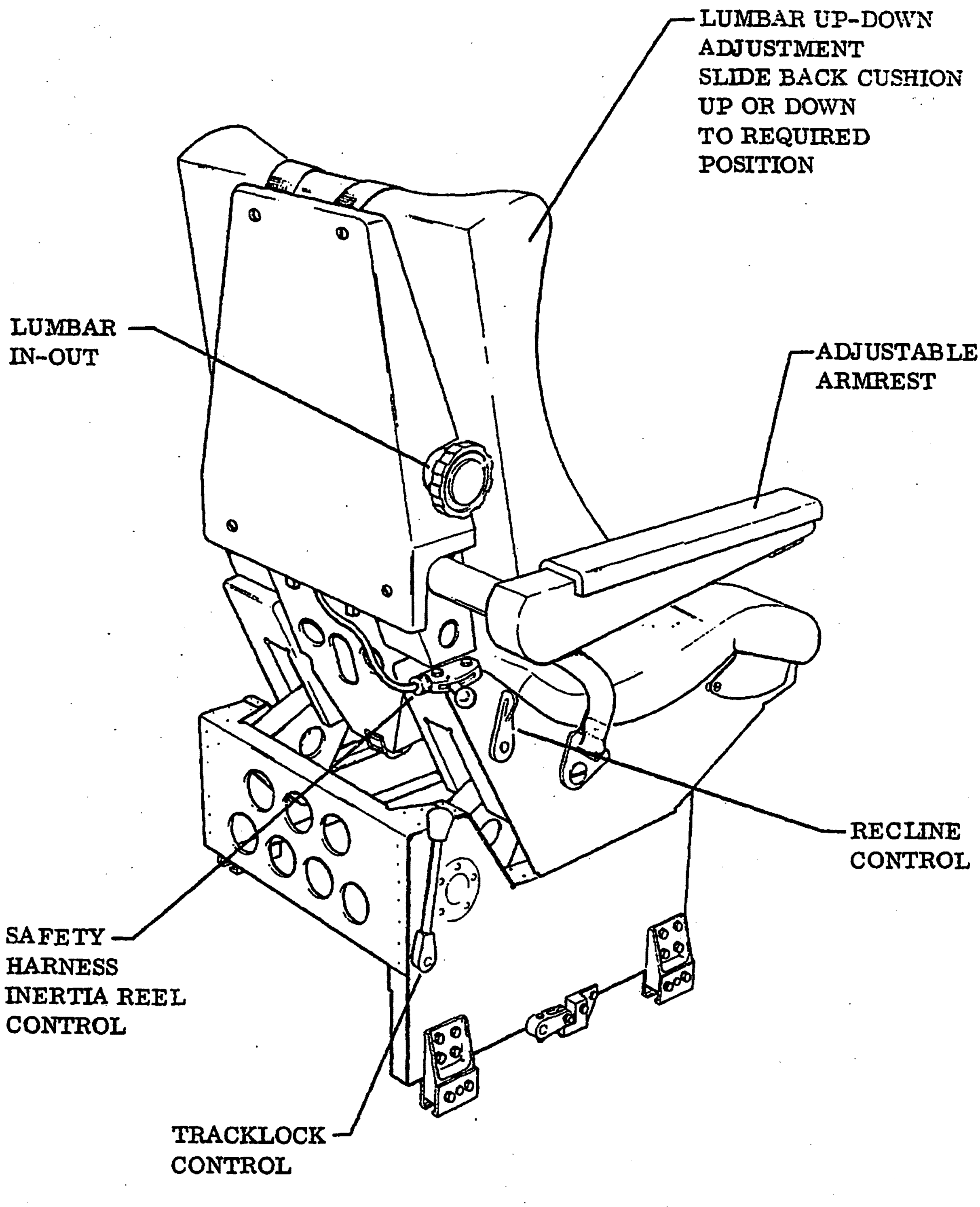
PILOT'S AND COPILOT'S SEATS - DESCRIPTION AND OPERATION

1. Description
   A. The pilot's and copilot's seats are manually adjustable in the forward, aft and vertical positions. Forward-aft adjustment is accomplished by a lever on the aft-inboard part of the seat. Vertical adjustment is accomplished by a lever located on the forward-outboard part of the seat. An optional armrest located on the inboard side of each seat is adjustable for ease of entry and exit from the seat. The metal structure seats are padded with foam rubber and upholstered.
   B. Each seat is equipped with a two-piece nylon seat belt which is fastened to the seat structure on the inboard side and the fuselage structure on the outboard side. An optional shoulder harness is also available. The lower portion of the shoulder harness is attached to the existing seat belt and the upper portion is attached to a reel mounted above each crew seat.
   C. Aircraft Equipped with optional IPECO Pilot's and Copilot's seats. (See figure 1.)
      (1) The crew seat is of an advanced ergonomic design providing maximum long term comfort while not restricting the free movement of the occupants. The seat is of lightweight construction and comprises two basic structures, the upper seat structure contains the necessary controls to adjust the seat; the seat base houses the positive track locking mechanism. The seat is designed to give long service life with the minimum of maintenance and overhaul.
      (2) The seat is either left or right, (pilot or copilot), depending upon the positioning of the major controls.
      (3) The seat base consists of a light alloy box structure with left and right hand claw plate and roller assemblies, running on aircraft mounted tracks. The side panels are asymetrically proportioned in height to allow for the contours of the aircraft structure. A positive lock mechanism allows fore and aft adjustment of the seat position; the mechanism is tensioned by springs.
      (4) The light alloy seat assembly contains the necessary controls and mechanism to adjust the seat pan height, recline, and lumbar support. The single inboard armrest of each seat is adjustable for height and can be folded and stowed when not in use.
      (5) The safety harness inertia reel is mounted at the rear of the spine; the lap straps and crotch strap are attached to the seat pan.

2. Operation
   A. Component Operation
      (1) The track locking mechanism is spring loaded by two tension springs. When the tracklock lever is pulled rearwards, the tracklock pins disengage from the track allowing the seat to be moved fore or aft to the required position.



Pilot's and Copilot's Seat
Figure 1 (Sheet 1 of 2)



Pilot's and Copilot's Seat
Figure 1 (Sheet 2 of 2)

(2) When the seat height adjustment handle is pulled upwards, height lock pins are disengaged from lock plates in the seat base. Continued pressure on the handle rotates the front and rear lift arms which are in turn assisted by spring tension of the bungee rubber cords to raise the seat. When the handle is released the locking pins engage the locking plates in the desired position. When the seat is to be raised or lowered, it is necessary for the occupant to adjust his weight on the seat pan until the required position is obtained.

(3) The thigh pads are spring loaded on pivot shafts with operating levers running in slides on the seat pan. When the seat occupant operates the aircraft controls, thigh pressure pushes the thigh pads down against spring tension. When pressure is relaxed, the pads return to the original position.

(4) The recline unit is mounted between the centre of the spine base and the underside of the seat pan. The recline control is situated at the rear of the seat pan. When the control handle is pulled upward, a spring loaded latch plate is withdrawn by a cable from contact with a coarse threaded nut within the recline unit. Application of pressure on the backrest rotates the threaded nut and allows the strut to be pushed into the recline unit; the spring is compressed and the backrest reclines. When the control handle is released, the spring loaded latch plate re-engages with the coarse threaded nut and the backrest is locked in the required position. If the control handle is again pulled upward, and pressure released from the back rest, the spring, which was compressed by the recline operation, re-asserts itself causing the threaded nut to rotate in the opposite direction. The strut pushes on the backrest which returns to an upright position and is locked upon release of the control handle.

(5) The armrest is individually adjustable for height by a control knob located at the forward end and connected by a drive shaft to a screw adjuster. When the control knob is turned, the screw adjuster increases or decreases the distance between the armrest drive plate and the pivot block, thereby raising or lowering the armrest. A separate fixed up stop makes it possible to raise the armrest from any position of adjustment to align parallel to the seat back. When fully folded, the armrest is pushed in towards the centre of the seat, thus reducing the seat width.

(6) The seat back cushion is adjustable for up-down and in-out lumbar support. Lumbar in-out adjustment is controlled by a handwheel on the inboard side of the seat. When the handwheel is rotated, the movement is transmitted through worm and wheel gears to the cross shaft, to which two relay arms are connected. As the relay arms rotate, the lumbar cushion moves forward or backward to the spine structure.

The up-down adjustment is achieved by sliding the backrest up or down to the required position, the weight of the back cushion being balanced by the spring housed in the seat spine.

(7) Remove the forward stops from the floor rails. Remove the rear claw plates from the seat base. Note the location of the claw plates fitted with the captive nuts for the assembly sequence. Place the rear claw plate assemblies on the floor rails by leading them on to the front of the rails. Position the seat with the front claw plate assemblies forward of the floor rails and draw the seat rearwards until the front claw plates and rollers locate fully on the rails. Secure the rear claw plates and rollers to the seat base.

PILOT'S AND COPILOT'S SEATS - MAINTENANCE PRACTICES

1. Removal/Installation

NOTE: • Removal of the pilot's and copilot's seats is identical.

• To facilitate removal and installation of the seats, it is recommended that the pilot seat be removed first and installed last.

A. Remove Seat (Aircraft Not Equipped with Optional IPECO Pilot's and Copilot's Seats.)
(1) Lower the seat as far as possible using the vertical adjustment lever.
(2) Remove seat cushion for ease of access to attaching hardware.
(3) Loosen nuts and bolts which pass through seat rail shoe. Nuts should be loosened sufficiently to remove seat from seat rail.

CAUTION: AVOID REPOSITIONING VERTICAL ADJUSTMENT LEVER WHILE REMOVING SEAT FROM COCKPIT. INADVERTENT MOVEMENT OF LEVER WILL CAUSE THE HYDROLOK ASSEMBLY TO TELESCOPE THE SEAT AND MAY CAUSE DAMAGE WITHIN THE COCKPIT.

(4) Spread shoes sufficiently to clear seat rail. Lift seat from rail and remove from aircraft.

B. Install Seat (Aircraft Not Equipped with Optional IPECO Pilot's and Copilot's Seats.)

CAUTION: AVOID REPOSITIONING VERTICAL ADJUSTMENT LEVER WHILE INSTALLING SEAT IN COCKPIT. INADVERTENT MOVEMENT OF LEVER WILL CAUSE THE HYDROLOK ASSEMBLY TO TELESCOPE THE SEAT AND MAY CAUSE DAMAGE WITHIN THE COCKPIT.

(1) Position seat over rails, spread shoes sufficiently to clear rail and lower seat onto seat rail.
(2) Tighten nuts and bolts securing shoes.
(3) Install seat cushion.
(4) Check seat operation through its full travel on the seat rail for positive pin engagement and release. Also check for binding of seat as it slides on rail.

C. Remove Pilot/Copilot Seat (Aircraft Equipped with Optional IPECO Pilot's and Copilot's Seats.)
(1) Remove forward seat track stops.

NOTE: Nuts are captive on inner plates. Retain spacers released when bolts are withdrawn.

(2) Remove two attach bolts from each rear claw foot, and remove claw feet.
(3) Hold tracklock lever in its rear most position and move seat forward until front claw feet disengage from seat tracks.
(4) Remove seat from aircraft.

D. Install Pilot/Copilot Seat (Aircraft Equipped with Optional IPECO Pilot's and Copilot's Seats.)
(1) Verify that forward seat track stops are removed from seat tracks and that rear claw feet are removed from seat.
(2) Position seat with front claw feet immediately forward of seat tracks.

NOTE: The seat must approach seat tracks from outboard side with tracklock lever held in the rear most position to allow the clamp plates to engage with the seat track heads.

(3) With tracklock lever in its rear most position, move seat aft until front claw feet engage seat tracks.
(4) Install rear claw feet, making sure that each bolt has a spacer installed.
(5) Check seat for freedom of movement on seat rails. Check that tracklock pins will engage all lock holes in seat tracks.
(6) Install forward seat track stops.

2. Adjustment/Test

A. Functional Test of Crew Seat Mechanisms (Aircraft Equipped with Optional IPECO Pilot's and Copilot's Seats)
(1) Sit in the seat and operate the tracklock lever. Check that the seat runs freely on the floor rails. Release the tracklock lever and check that the locking pins engage freely in the holes in the floor rail.
(2) Operate the seat height adjustment lever and check that the seat moves freely over its full range and that it locks when the lever is released.
(3) Push the thigh pads down and check that the override machanism operates smoothly. Relax pressure and check that the thigh pads return to their pre-set position.
(4) Operate the armrest adjuster and check that the armrest can be raised or lowered over the full range. Fold back the armrest and check that it can be pushed inwards to the stowed position and withdrawn again.
(5) Operate the lumbar support in-out handwheel and check that the lumbar cushion moves freely through its full range of movement.
(6) Slide the backrest fully up and down through its full range and check that it remains in any selected position.

3. Inspection/Check

A. Check all seat parts for the following:

(1) Cleanliness.
(2) Distortion.
(3) Cracks - visually and dye penetrant.
(4) Scores.
(5) Dents.
(6) Excessive Wear.
(7) Corrosion and deterioration of protective treatment.
(8) Serviceability of all screw threads.
(9) Security of attachment.
(10) Locking quadrant holes for wear and distortion.
(11) Fraying of height lock cable.
(12) General condition of bungee cords.

B. Check all rivets and riveted assemblies for the following:

(1) Tightness.
(2) Flush fitting where applicable.
(3) Security of attachment.
(4) Distortion of holes.

4. Approved Repairs

A. General

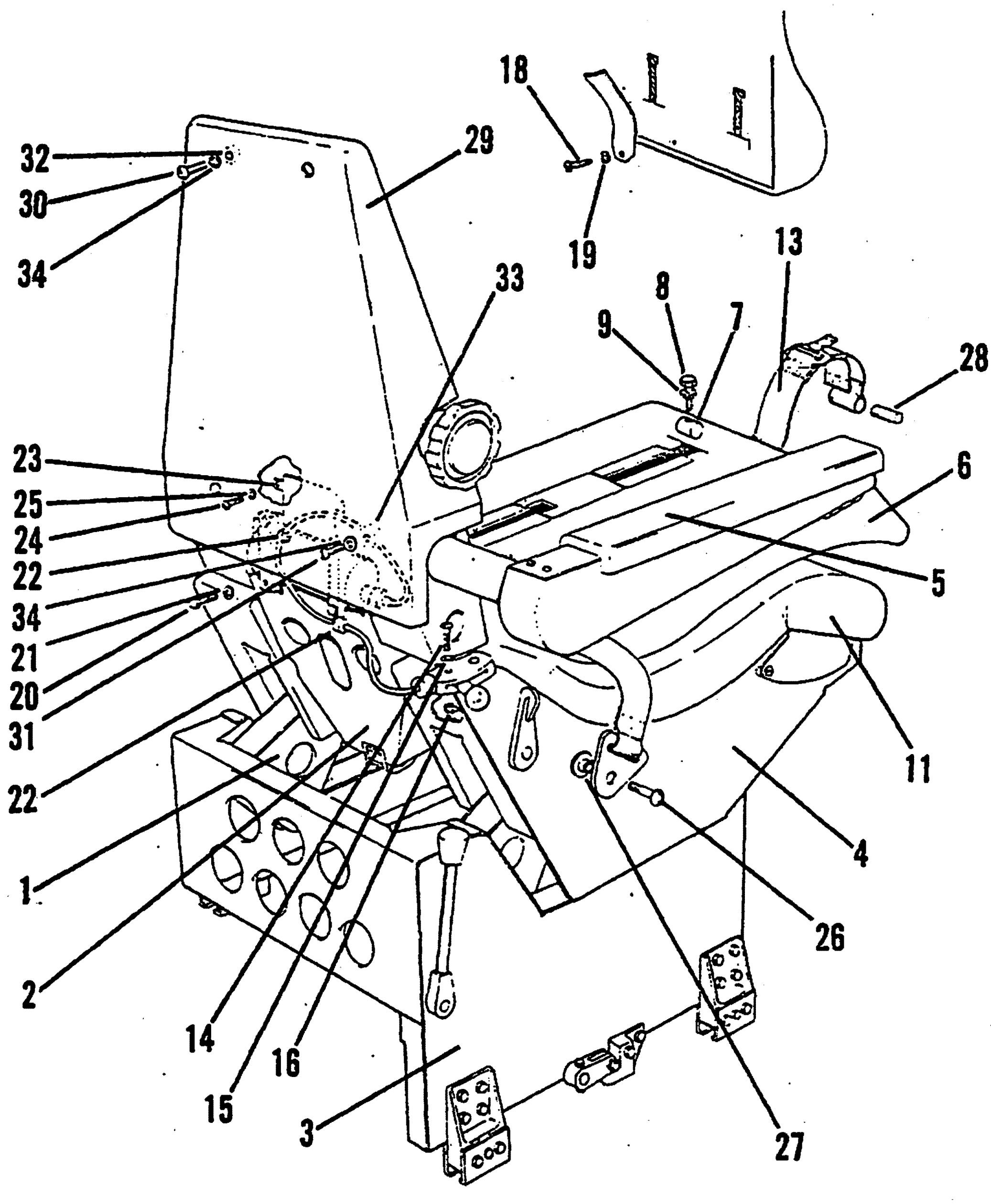
(1) Repairs to the seat are confined to the replacement of parts as required or welded assemblies are to be returned to IPECO for repair or replacement.

WARNING: UNAUTHORIZED REPAIRS TO SEAT STRUCTURE OR COMPONENTS MAY AFFECT THE SAFETY FEATURES OF THE SEAT AND MUST NOT BE ATTEMPTED.

NOTE: If damage has occurred to major assemblies of the seat, e.g. the spine assembly, seat pan, or base assembly, the complete seat should be returned to the manufacturer for repair.

B. Disassemble Upholstery and Accessories (See figure 201.)

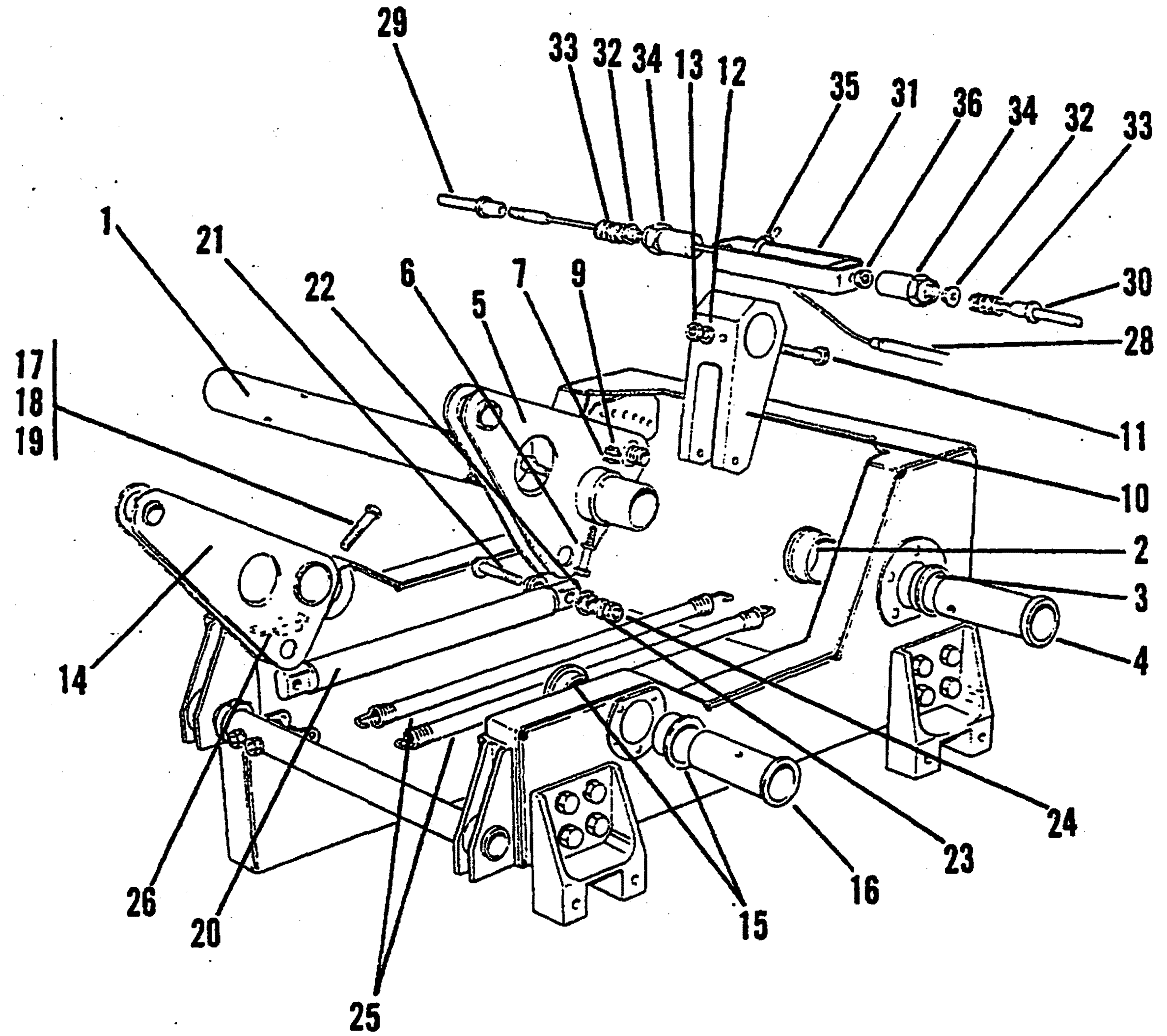
(1) Remove bolts (26) and remove lap straps. Retain spacers (27).
(2) Push crotch strap (13) back into its mounting bracket until shaft (28) can be removed.
(3) Remove bolts (30 and 31), washers (34) and remove back fairing (29). Retain spacers (32 and 33).
(4) Remove bolts (24), washers (25), and remove harness fairing (23).
(5) Remove bolts (20 and 14), washers (21 and 15), and nut (16) and remove inertia reel and release assembly complete. Retain clips (22).



Upholstery and Accessories
Figure 201

(6) Remove seat pan cushion (11) by pulling upwards from rear to release Velcro attachment strips.
(7) Remove bolts (8), washers (9), and remove stop block (7).
(8) Remove bolt (18) and washer (19) to disconnect tensator spring. Remove seat back rest assembly (6) by moving downward to clear rollers on spine assembly.
(9) Refer to Figure 206 and remove screws (20) and washers (21) securing cover (19). Remove bolts and washers (17) and withdraw armrest (16) assembly from armrest tube. Retain spacers (18).

C. Disassemble Seat Base and Lift Mechanism (See figure 202.)
(1) Refer to figure 205 and remove bolts (48), nuts (50), and washers (49) from each pivot tube (47).
(2) Refer to figure 205 and remove Sel-loc pins (54) from spacer (53).
(3) Refer to figure 205 and remove pivot tube (47) while supporting front of seat pan. Retain bearings (51).
(4) Refer to figure 205 and remove bolts (61), nuts (63), and washers (62) from each rear lift arm.
(5) Refer to figure 205 and remove pivot pins (60) while supporting rear of seat pan. Retain bushings (59).
(6) Remove bolts (21), nuts (24), washers (22), bushings (23), and remove operating bars (20). Retain bushings (23).
(7) Remove bolts, nuts and washers (17), (18), (19), and remove pivots (16). Retain bearings (15) and front lift arms (14).
(8) Remove bungee rubber cords (25).
(9) Remove bolt (11), nut (13), and washer (12).
(10) Remove bolts (6), nuts (9), and washers (7) from rear lift arms (5).
(11) Unscrew spring housings (34), remove cable tie (35) and disconnect cable assembly (28) from connector bracket.
(12) Unscrew pin (29) from cable assembly (28) end fitting. Remove ferrule (32) and spring (33).
(13) Unscrew pin (30). Remove ferrule (32), spring (33) and locknut (36).
(14) Remove torque tube (1) from base. Retain bungee lever (10), lift arms (5), sleeve tubes (4), and bushings (2 and 3).
(15) Refer to figure 205 and remove bolt (73) and washer (74) from 'P' clip (72). Remove height lock cable assembly (71).
(16) Refer to figure 205 and drive out Sel-lok pins (66), remove pivot pins (65) and recline unit (64) from seat.
(17) Refer to figure 205 and drive out Sel-lok pin (34) and detach lever (33) from recline shaft (28).
(18) Refer to figure 205 and remove recline shaft (28) from seat. Retain bearings (29 and 30).
(19) Refer to figure 205 and drive out Sel-lok pin (32) and remove recline lever (31) from shaft (28).



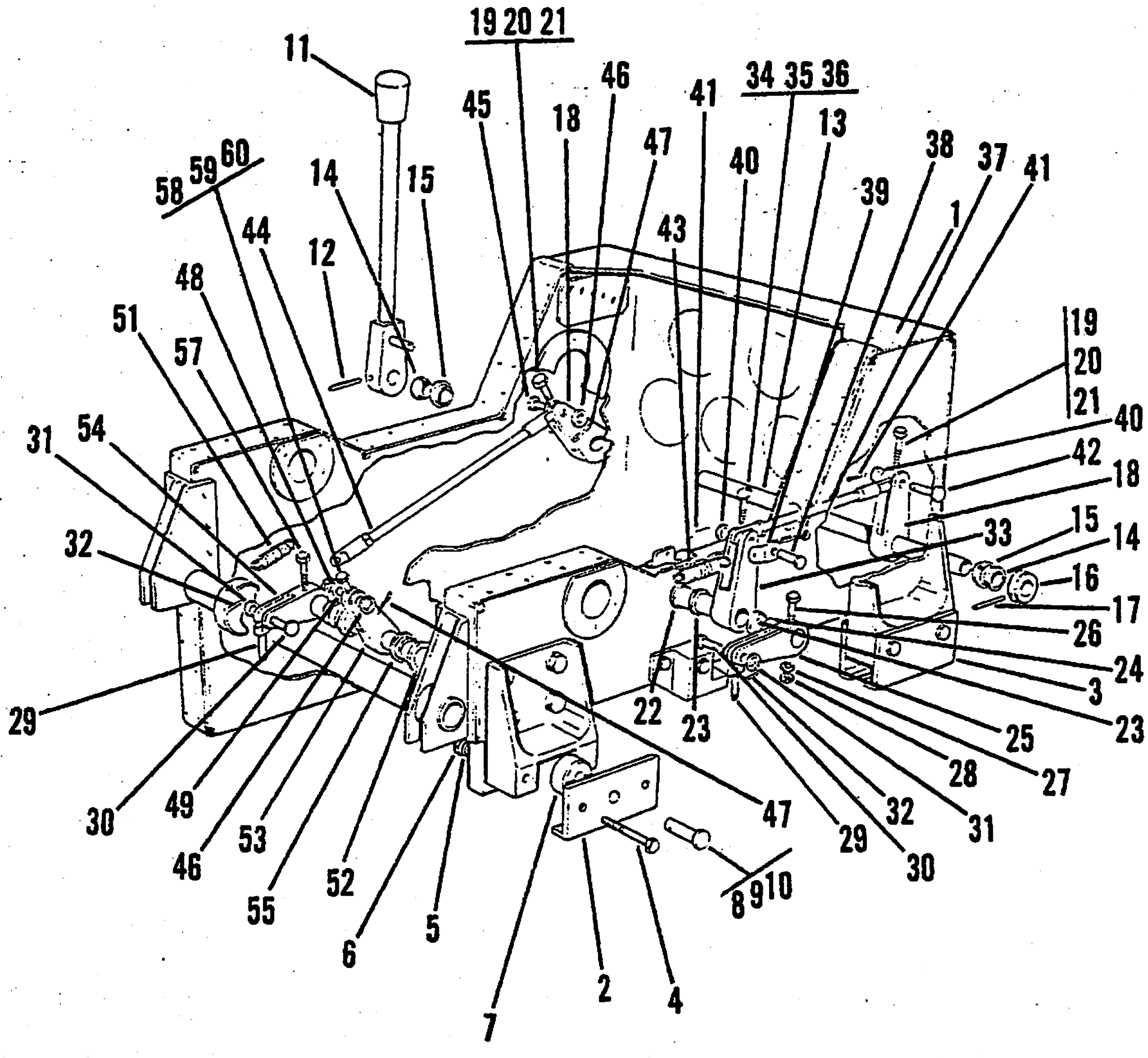
Seat Base and Mechanism
Figure 202

D. Disassemble Seat Base and Tracklock Mechanism (See figure 203.)

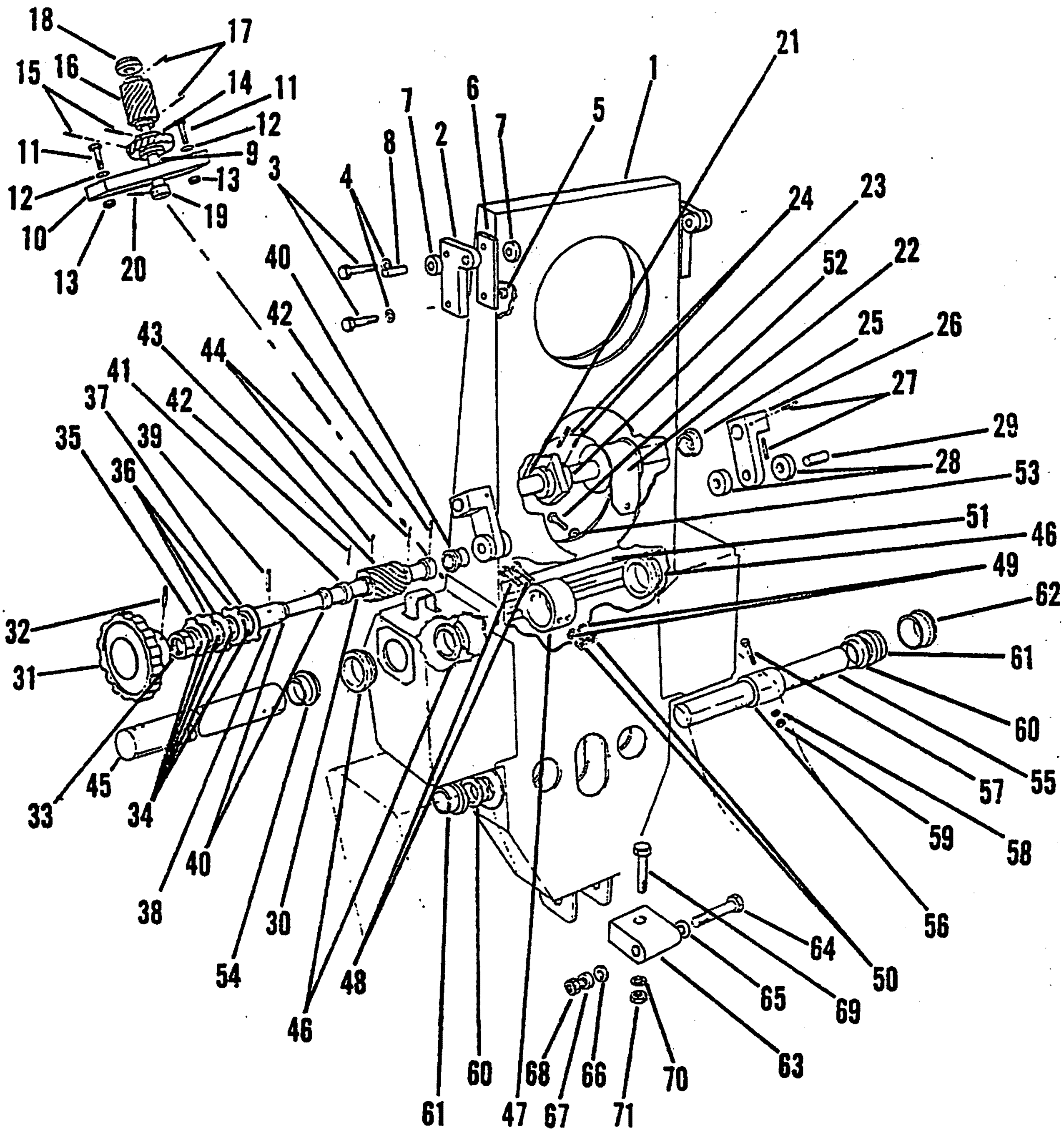
(1) Drive out spirol pin (12) and remove handle assembly (11) from shaft (13).
(2) Remove tension springs (51 and 43).
(3) Remove shear pin (38) from lever (33). Retain split pin (41), washer (40), and spring link (39).
(4) Remove pin (48) from lever (53). Retain split pin (47), washer (46), and spring link (49).
(5) Remove bolt (19), nut (20), and washer (21) from lever (18) outboard.
(6) Remove bolt (19), nut (20), and washer (21) from lever (18) inboard.
(7) Remove split pin (41) and shear pin (42). Retain washer (40). Remove control rod (37) from base.
(8) Remove split pin (47) and shear pin (45). Retain washer (46) and remove control rod (44) from base.
(9) Remove shaft (13). Retain bearing housing (14) and bearing (15).
(10) Drive out Sel-lok pin (17) and remove collar (16) from shaft (13).
(11) Remove bolt (26), nut (28), and washer (27) from lever (25).
(12) Remove bolt (34), nut (35), and washer (36) from lever (33).
(13) Remove pivot tube (22) from base. Retain levers (25 and 33), bushing (23) and spacer (24).
(14) Remove split pin (32) and shear pin (30). Retain washer (31). Remove track lock pin (29).
(15) Remove bolt (58), nut (59), and washer (60) from lever (54).
(16) Remove bolt (60), nut (58), and washer (59) from lever (53).
(17) Remove pivot tube (52). Retain bushing (55), levers (53 and 54), and spacer (56).
(18) Remove split pin (32) and clevis pin (30). Retain washer (31) and tracklock pin (29).
(19) Remove bolts (4), nuts (6), washers (5), and claw plates (2 and 3) from base.
(20) Remove shear pins (8), washers (9), and split pins (10) from each claw plate assembly. Retain cam follower (7).

E. Disassemble Seat Spine and Mechanism (See figure 204.)

(1) Remove spindle (8) and rollers (7) from spine.
(2) Remove bolts (3) and washers (4) and detach pivot plates (2) and packers (6).
(3) Remove spindle (29) and rollers (28).
(4) Remove Sel-lok pins (27) and detach lumbar levers (26) from shaft (21).
(5) Slide shaft (21) until the worm wheel (23) is clear of the vertical shaft assembly. Drive out Sel-lok pins (24).
(6) Remove shaft (21). Retain worm wheel (23), spring tensator (22), and bearings (25).
(7) Remove bolts (11), nuts (13), and washers (12) and remove vertical shaft assembly (9) from spine. Retain bearing (18).



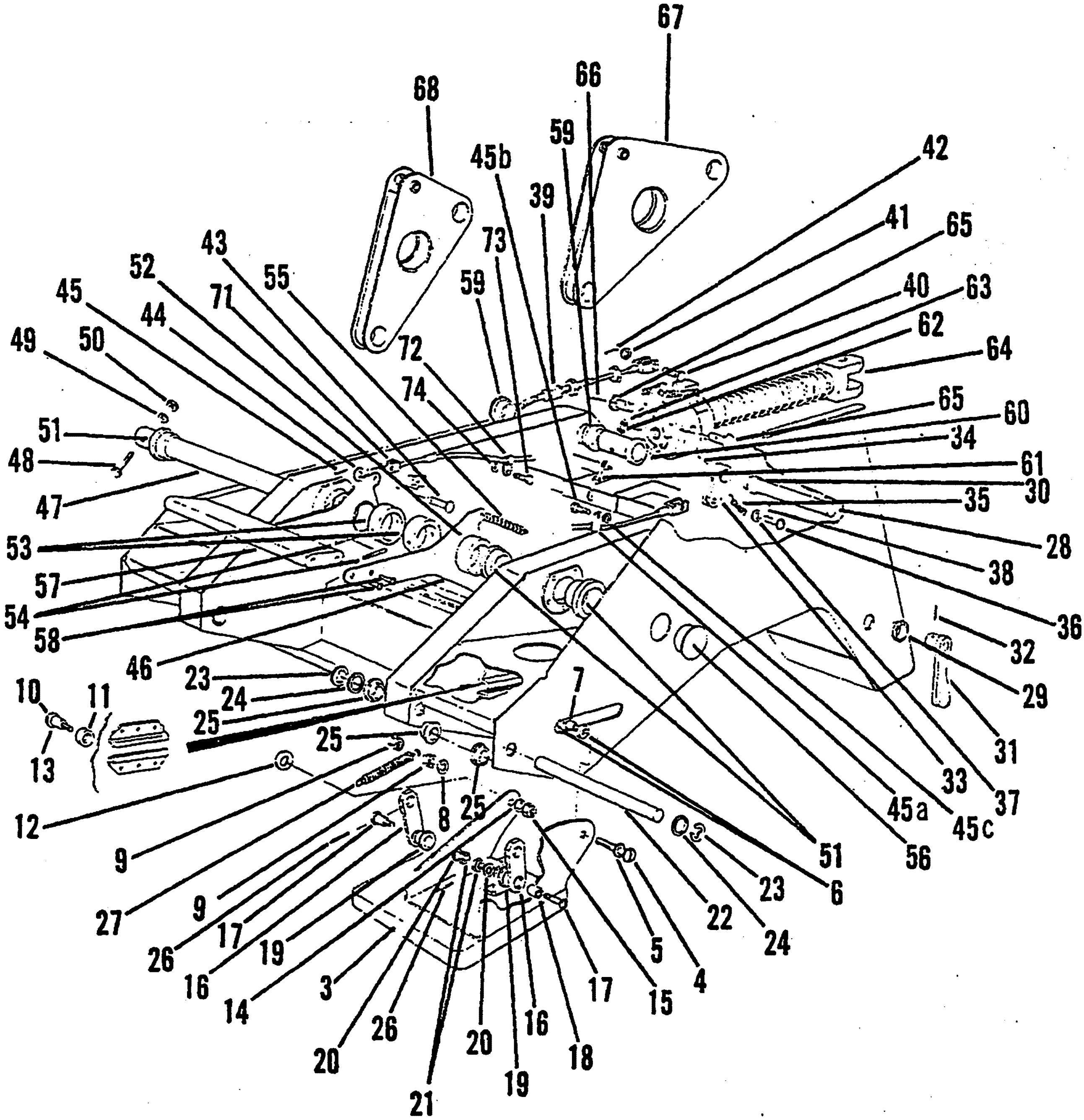
Seat Base and Track Lock Mechanism
Figure 203



Seat Spine and Mechanism
Figure 204

(8) Drive out Sel-lok pins (17) and remove worm (16) from shaft.
(9) Drive out Sel-lok pins (15) and remove worm wheel (14) from shaft.
(10) Drive out Sel-lok pin (20) and remove collar (19).
(11) Separate shaft (9) from bearing block (10).
(12) Drive out Sel-lok pin (32) and remove handwheel (31). Remove locknut (33.)
(13) Remove catch plate washers (34), catch plates (35 and 36), and stop plate (37) from drive-shaft (30).
(14) Drive out Sel-lok pin (39) and remove adapter (38).
(15) Drive out Sel-lok pins (44) from worm wheel (43).
(16) Drive out Sel-lok pins (42) from sleeve (41).
(17) Remove shaft from spine. Retain all parts mounted on the shaft including bearings (40).
(18) Remove bolts (48), nuts (50), and washers (49).
(19) Remove armrest tube from spine. Retain armrest tube stop (47) and bearings (46).
(20) Remove bolts (52) and washers (53) and detach stop block (51) from the spine.
(21) Remove bolt (57), nut (59), and washer (58) from pivot shaft collar (56).
(22) Remove bolt (64), nut (68), and washers (65, 66, and 67) from attachment block (63).
(23) Support spine assembly and remove spline shaft (55). Retain bearings (60) and pivot bearing (61), and collar (56).
(24) Remove bolt (69), nut (71), and washer (70) and remove block (63) from recline unit.

F. Disassemble Seat Pan and Mechanism (See figure 205.)
(1) Remove bolts (61), nuts (63), and washers (62). Remove pivot pin (60) and rear lift arms.
(2) Remove shear pin (43), washer (44), and split pin (45) and disconnect height lock cable (71) from height lock lever (46).
(3) Remove bolts (48), nuts (50), and washers (49) and remove forward lift arms (68).
(4) Remove end plug (56).
(5) Drive out Sel-lok pins (54) from spacers (53) and withdraw pivot tube (47). Retain height lock lever assembly (46), bearings (51 and 52), and spacers (53). Detach spring (55).



Seat Pan and Mechanism
Figure 205

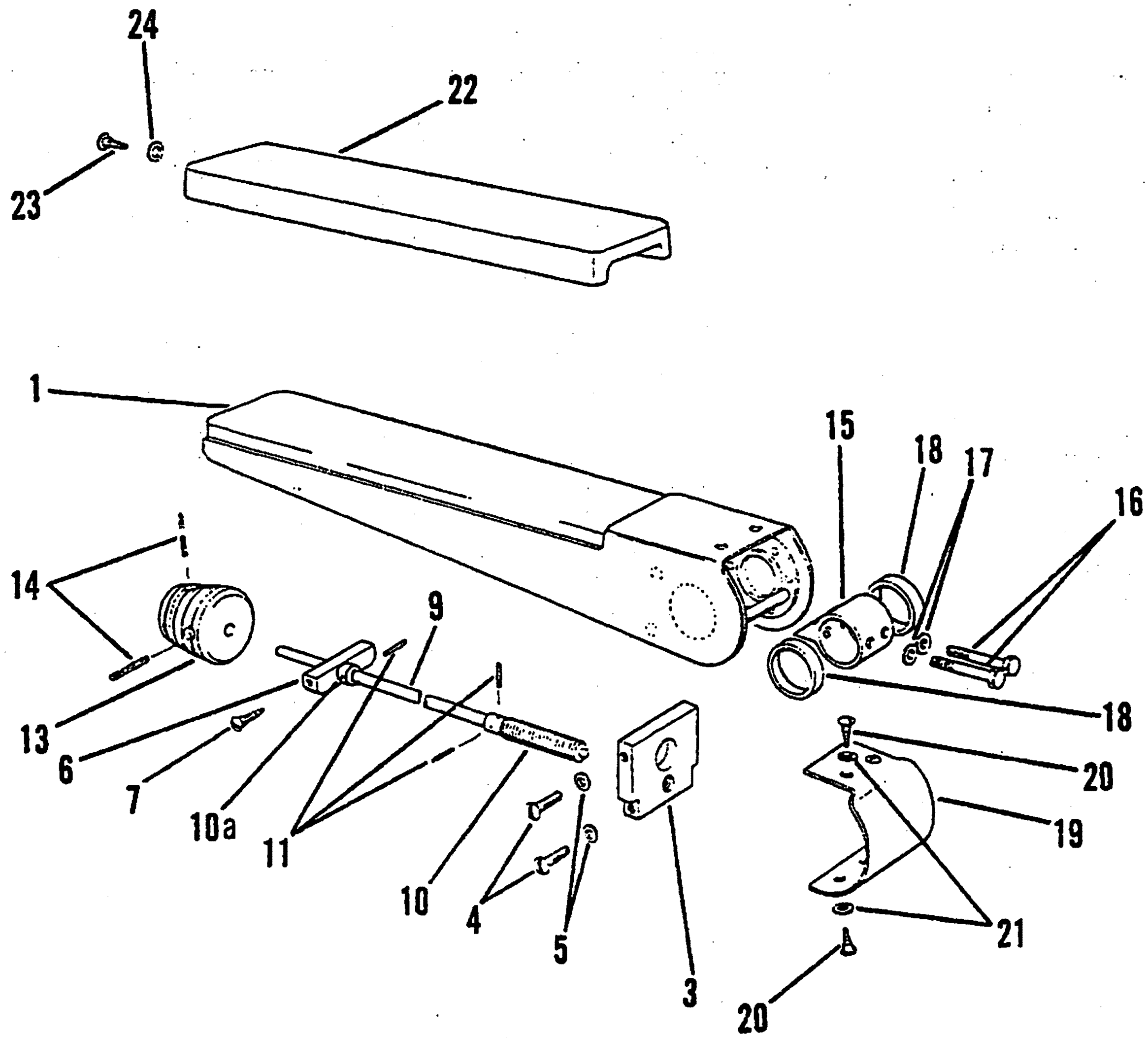
EFFECTIVITY: OPTIONAL

(6) Remove screws (58) and separate handle (57) from height lock levers (46).
(7) Drive out Sel-lok pins (26) from levers (16).
(8) Remove circlips (23) and washers (24) and remove shafts (22). Retain bearings (25).
(9) Remove bolts (4 and 10), nuts (9 and 15). Unhook tension spring (27), washers (5, 14, and 8). Remove thigh pads (3). Retain rollers (6 and 11), sleeves (7, 18, and 13), and washer (12).
(10) Remove bolts (17), nuts (21), washers (20), sleeves (18) and detach levers (16). Retain bearings (19).
(11) Unhook tension spring (35). Remove pin (36), washer (37), and split pin (38) to release recline cable from lever (33).
(12) Remove pin (40), washer (41), and split pin (42) and disconnect recline cable (39) from recline unit.
(13) Remove bolts (45b) and washers (45c) and release 'P' clips (45a) from structure. Remove recline cable assembly (39).

G. Disassemble Armrest (See figure 206.)
(1) Remove screw (23) and washer (24) and remove capping (22).
(2) Drive out Sel-lok pins (14) and remove adjusting knob (13).
(3) Screw shaft assembly (9) fully rearward until it is disconnected from threaded block (3).
(4) Drive out Sel-lok pins (11) and separate drive shaft (9) and adjustment screw (10).
(5) Remove screws (4) and washers (5) and remove block (3) from armrest.
(6) Remove drive screws (7) and bearing block (6).

NOTE: In order to perform steps G (5) and (6), the covering material must be released from the structure, and refitted, using a suitable adhesive, during re-assembly.

H. Assemble Base and Tracklock (See figure 203.)
(1) Assemble bearing housing (14) and bearing (15) to base assembly.
(2) Assemble operating handle (11) to shaft (13) using new Sel-lok pin (12).
(3) Insert shaft assembly (13) into base attaching each lever (18) in turn while the shaft is inserted.
(4) Secure each lever (18) to shaft with bolts (19), nuts (20), and washers (21).
(5) Secure collar (16) to shaft (13) with new Sel-lok pin (17).
(6) Insert control rod (37) into base and secure to lever (18) with shear pin (42), washer (40), and split pin (41).
(7) Insert control rod (44) into base and secure to lever (18) with shear pin (45), washer (46), and split pin (47).
(8) Fit bushing (23) to base.



Armrest Assembly
Figure 206

(9) Assemble levers (25 and 33) to base with spacer (24), bushing (23), and pivot tube (22).
(10) Secure lever (33) using bolt (34), nut (35), and washer (36).
(11) Secure lever (25) using bolt (26), nut (28), and washer (27).
(12) Attach control rod (37) to lever (33) using shear pin (38), link (39), washer (40), and split pin (41).
(13) Attach tracklock pin (29) to lever (25) using clevis pin (30), washer (31), and split pin (32).
(14) Assemble bushing (55) to base.
(15) Assemble levers (53 and 54) with spacer (56) and pivot tube (52).
(16) Secure lever (54) with bolt (57), nut (59), and washer (58).
(17) Secure lever (53) with bolt (58), nut (60), and washer (59).
(18) Attach control rod (44) to lever (53) with shear pin (48), spring link (49), washer (46), and split pin (47).
(19) Attach tracklock pin (29) to lever (54) with pin (30), washer (31), and split pin (32).
(20) Attach springs (51 and 43) to spring links and anchor points.
(21) Assemble cam follower (7) to claw plates with pins (8), washers (9), and split pins (10).
(22) Attach claw plates (2 and 3) to base using bolts (4), nuts (6), and washers (5). Note that each rear claw plate (3) is fitted on the outboard side.

I. Assemble Seat Pan and Mechanism (See figure 205.)
(1) Assemble levers (16) to thigh support brackets (3) with bolts (17), sleeves (18), bearings (19), nuts (21), and washers (20).
(2) Attach thigh support brackets to seat pan using bolts (4 and 10), sleeves (13 and 7), rollers (6 and 11), washers (5, 14, 8, and 12), and nuts (9 and 15).
(3) Attach levers (16) to seat using bearings (25), shaft (22), washers (24), and circlips (23).
(4) Secure levers to shaft with new Sel-lok pins (26).
(5) Attach springs (27).
(6) Secure operating handle (31) to tube (28) with new Sel-lok pin (32).
(7) Fit bearings (29 and 30) and tube (28) to seat.
(8) Secure lever (33) to tube using new Sel-lok pin (34).
(9) Secure recline unit (64) to seat using pins (65) and new Sel-lok pins (66).
(10) Attach recline cable assembly (39) to recline unit using pin (40), washer (41), and split pin (42).

J. Assemble Seat Base and Pan Assembly (See figure 202.)
(1) Assemble bearings (2 and 3) and pivot tubes (4) to base.
(2) Assemble torque tube (1) to base with rear lift arms (5), and bungee lever (10).
(3) Secure bungee lever to tube using bolt (11), nut (13), and washer (12).
(4) Secure rear lift arms (5) to torque tube using bolts (6), nuts (9), and washers (7).

(5) Fit bearings (15) and pivot tubes (16) to base and fit front lift arms (14).
(6) Secure lift arms using bolts (17), washers (18), and nuts (19).
(7) Assemble bearings (22) to lift arms. Install operating bars (20) and secure with bolts (21), nuts (23), and washers (22).
(8) Assemble pin (30), spring (33), ferrule (32), to housing (34) and secure assembly with locknut (36) to connector bracket (31). Attach assembly to rear outboard lift arm.
(9) Assemble ferrule (32), spring (33), housing (34), and pin (29) to cable assembly.
(10) Engage cable end out casing in connector bracket and screw spring housing on to inboard rear lift arm.
(11) Fit cable tie (35) around connector bracket and cable assembly.
(12) Attach bungee rubber cords to bungee lever and forward cross tube.
(13) Refer to figure 205 and assemble bearings (59) to seat pan.
(14) With the front of the seat pan positioned at the front lift arms, assemble height lock levers (46), collars (53), spacers (52), torque tube (47), and seat pan to base.
(15) Assemble spacers (59), figure 205, to base.
(16) Assemble seat base to rear lift arms with pivot tubes (60).
(17) Secure lift arms to seat base using bolts (48 and 61), nuts (50 and 63), and washers (49 and 62).
(18) Attach height lock handle (57), figure 205, to height lock levers (46) using screws (58).
(19) Attach height lock cable (71) to height lock levers (46) using pin (43), washer (44), and split pin (45).
(20) Refer to figure 205 and attach height lock cable to structure using 'P' clip (72), bolt (73), and washer (74).
(21) Refer to figure 205 and attach tension spring (55).
(22) Attach free end of recline cable (39) to lever (33) using pin (36), washer (37), and split pin (38).
(23) Refer to figure 205 and secure recline cable to seat structure with 'P' clips (45a), bolts (45b), and washers (45c).

K. Assemble Seat Spine and Mechanism (See figure 204.)
(1) Fit bearings (40) to spine structure. Assemble worm wheel (43), shaft (30) and collars (41) to spine. Secure collars and worm with new Sel-lok pins (42 and 44).
(2) Assemble bearings (25) to spine.
(3) Assemble shaft (21), tensator spring (22), lumbar levers (26), and worm wheel (23) to spine. Secure items (26) and (23) with new Sel-lok pins (24 and 27).

(4) Assemble worm wheels (14 and 16) to shaft (9) with new Sel-lok pins (15 and 17).
(5) Locate shaft assembly on to block (10) and attach collar (19) to shaft with new Sel-lok pin (20).
(6) Assemble completed assembly with bearing (18) to spine. Secure with bolts (11), nuts (13), and washers (12).
(7) Attach spacer (38) to shaft (30) with new Sel-lok pin (39). Assemble catch plates (34, 35, 36 and 37) to shaft.
(8) Install locknut (33) to shaft and install handwheel (31) to shaft with new Sel-lok pin (32). Tighten locknut.
(9) Install bearings (46) to spine.
(10) Secure guide block (51) to spine with bolts (52) and washers (53).
(11) Assemble armrest tube (45) to spine with stop (47). Secure stop to tube with bolts (48), nuts (50), and washers (49).
(12) Install block (54) to tube.
(13) Install spine assembly on seat pan assembly and assemble tube (55), spacer (56), and bearings (60 and 61). Install end plugs (62).
(14) Secure spacer (56) with bolt (57), nut (59), and washer (58).
(15) Assemble spine base to recline unit with block (63), bolts (64 and 69), washers (70, 66, and 67), and nuts (68 and 71).
(16) Secure pivot plates (2) with bolts (3), washers (4), and packer (6).

L. Assemble Armrest Assembly (See figure 206.)
(1) Install bearing block (6) to armrest with drive pins (7).
(2) Install block (3) to armrest with screws (4) and washers (5).
(3) Install shaft (9) and adjustment screw (10) with new Sel-lok pins (11).
(4) Install shaft assembly in bearing block (6), screw shaft into block (3).
(5) Secure knob (13) to shaft with new Sel-lok pins (14).
(6) Secure capping strip (22) with screw (23) and washer (24).

M. Upholstery and Accessories (See figure 201.)
(1) Assemble armrest to spine with spacers (18) and block (15). Secure with screws (16) and washers (17).
(2) Install cover (19) and secure with screws (20 and 21).
(3) Refer to figure 204 and install rollers (7 and 28) on spine with pins (8 and 29).
(4) Install backrest assembly on to spine and attach tensator spring to backrest with bolt (18) and washer (19).
(5) Install stops (7) with bolts (8) and washers (9).
(6) Install inertia reel to spine using bolts (20) and washers (21). Attach harness cable with 'P' clips under appropriate bolt heads.
(7) Secure harness fairing (23) to spine with bolts (24) and washers (25).

(8) Attach inertia reel operating handle to seat pan with bolts (14), nuts (16), and washers (15).
(9) Attach back fairing (29) to spine using bolts (30 and 31), washers (34), and spacers (32 and 33).
(10) Secure lap straps to seat pan using bolts (26) and spacers (27).
(11) Secure crotch strap to seat pan with shaft (28).
(12) Install seat cushion on seat pan.

PASSENGER COMPARTMENT - DESCRIPTION AND OPERATION

1. General

   A. The passenger compartment is that area aft of the service cabinet extending to the aft pressure bulkhead.

   B. Equipment and furnishings within the passenger compartment include the seats, toilet, service cabinets, life vests and a work table.

PASSENGER COMPARTMENT - MAINTENANCE PRACTICES

1. General
   A. The aircraft interior should be frequently cleaned to maintain its appearance and minimize deterioration.

2. Tools and Equipment

NAME	NUMBER	MANUFACTURER
Orvus WA Paste		
Lissapol	1132	Imperial Chemical Industries
Wich Torlan		Wichita Brush and Chemical Co.
Mineral Spirits	Apco No. 10	
Cheesecloth		Commercially Available

3. Cleaning Interior

CAUTION: DO NOT USE ABRASIVE CLEANERS, PARTICULARLY THOSE CONTAINING BLEACHES ON ANY INTERIOR DECORATIVE SURFACE. ABRASIVE CLEANERS WILL DESTROY SILK-SCREENED DESIGNS, DAMAGE COLORED LEATHERS, PAINTED SURFACES AND PLASTIC SURFACES. AVOID USING ORGANIC CHEMICALS, PARTICULARLY KETONES, ESTERS, AROMATIC AMINES, NITRO COMPOUNDS AND HYDROCARBONS OR CLEANERS WHICH MAY CONTAIN THE ABOVE CHEMICALS. AVOID USING MATERIALS SUCH AS FIRE EXTINGUISHER FLUIDS, DE-ICING FLUIDS, PAINT STRIPPERS, LACQUER THINNERS, OR HYDRAULIC FLUIDS.

   A. Cleaning Rugs, Leather, Fabrics
      (1) Mix 2 ounces of Orvus past per gallon of water.
      (2) Use only the foam for cleaning.
      (3) Sponge foam on the surface. Do not get surface too wet.
   B. Cleaning Plastics or Metal Surfaces

   CAUTION: DO NOT USE THE FOLLOWING MATERIALS ON DECORATIVE PLASTICS: GASOLINE, ALCOHOL, BENZENE, HEXANE, KEROSENE, XYLENE, ACETONE, CARBON TETRACHLORIDE, FIRE EXTINGUISHER OR DE-ICING FLUIDS, LACQUER THINNERS, OR WINDOW CLEANING SPRAYS.

(1) Mix 1 tablespoon Orvus paste per gallon of water.
(2) Wash the surface with a cloth.
(3) Rinse the surface with clean water and wipe dry.

C. Cleaning Acrylic Windows

(1) Remove loosely adhering dirt and grit from the window by flushing with water filtered free of dirt and abrasive materials.
(2) Wash with nonabrasive soap and water. A soft, thoroughly clean cloth, sponge, or chamois may be used in washing, but only as a means of carrying the soapy water to the plastic. Go over the surface only with the bare hand, so that any abrasive can be quickly detected. and removed before it scratches the plastic surface.

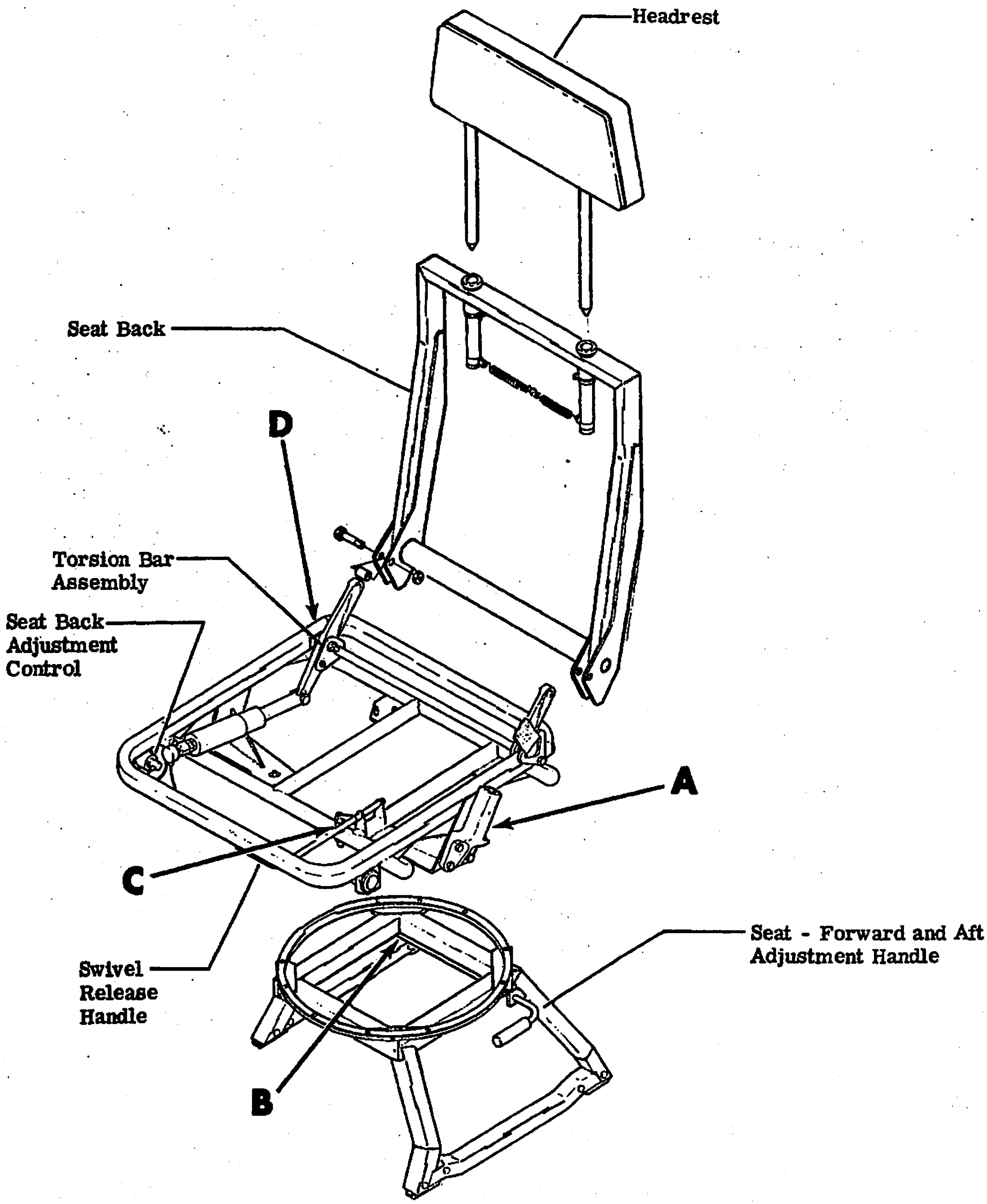
CAUTION: RUBBING THE PLASTIC SURFACE WITH A DRY CLOTH WILL CAUSE SCRATCHES AND BUILD UP AN ELECTROSTATIC CHARGE WHICH ATTRACTS DUST PARTICLES. ALL RUBBING OPERATIONS ON ACRYLIC PLASTICS SHALL BE DONE WITH AS LIGHT A PRESSURE AS POSSIBLE.

(3) Dry the window with a clean, damp chamois. A clean, soft cloth or tissue may be used if care is taken not to rub the plastic after it is dry.
(4) Remove oil and grease by rubbing lightly with a cloth wetted with aliphatic naphtha.

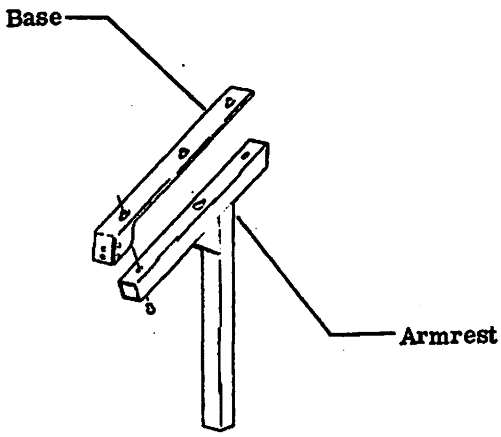
CAUTION: DO NOT USE THE FOLLOWING MATERIALS ON ACRYLIC PLASTICS: GASOLINE, ALCOHOL, BENZENE, HEXANE, XYLENE, ACETONE, CARBON TETRACHLORIDE, FIRE EXTINGUISHER, OR DE-ICING FLUIDS, LACQUER THINNERS, OR WINDOW CLEANING SPRAYS, AS THEY SOFTEN THE PLASTIC AND/OR CAUSE CRAZING.

PASSENGER SEATS - DESCRIPTION AND OPERATION

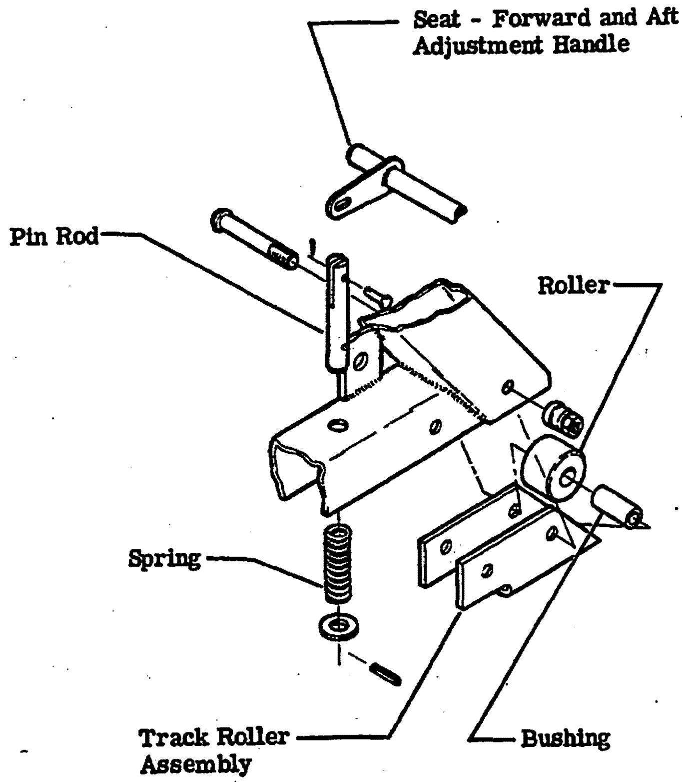
1. General
   A. Passenger seats in the aircraft consist of three types, a divan seat, a berthable divan and swivel seats.
   B. On 25B/D Aircraft, the divan seat is full cabin width located in the aft passenger compartment. The seat has split type seat backs which fold down against the seat bottom to provide additional cargo space.
   C. On 25C/F Aircraft, the berthable divan is full cabin width located in the aft passenger compartment. The seat has split type seat backs which may be adjusted by depressing a button located on the lower forward portion of the seat and locked in position by releasing the button. The berthable divan is designed to allow the seat backs to fold down into a bed. The divan also incorporates a center fold down arm rest.
   D. The swivel seats are mounted on seat rails which allow the seats to be moved fore and aft. The seats will swivel 180^{o} by releasing the swivel release and may be locked in the desired position. The seat may be reclined by depressing the button located on the lower forward portion of the seat and locked to the desired position by releasing the button. A telescoping head rest in the seat back is adjustable by pulling up and pushing down.



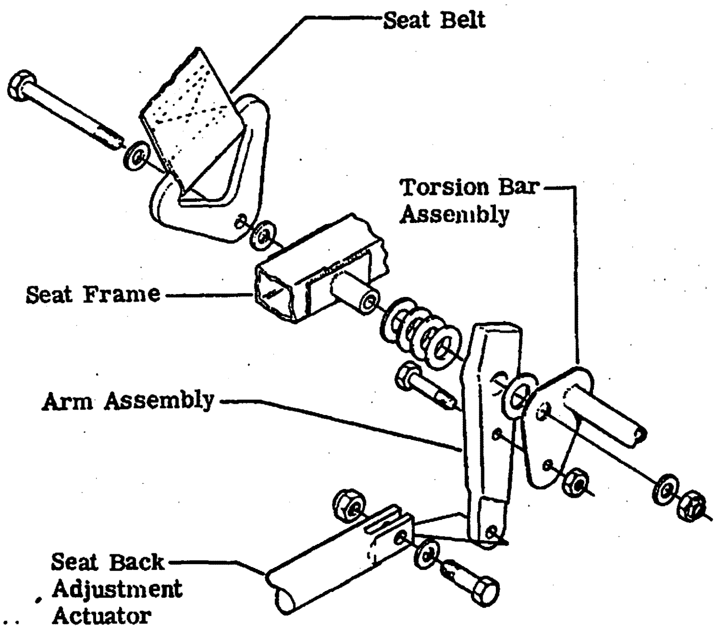
Swivel Seat Assembly
Figure 1 (Sheet 1 of 2)



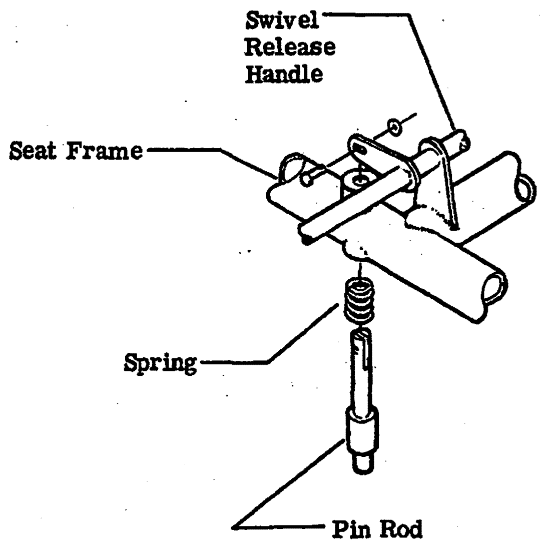
Detail A



Detail B

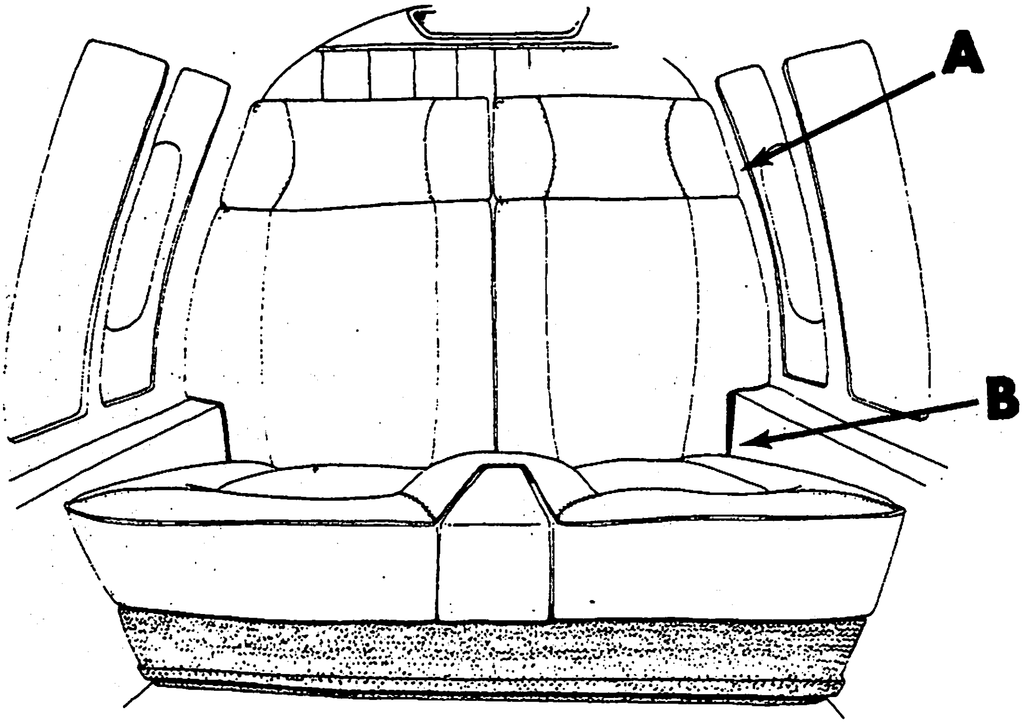


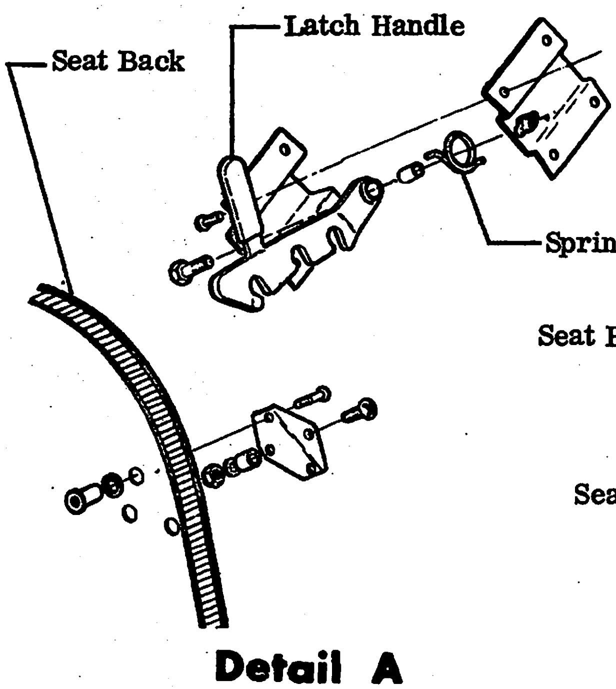
Detail D

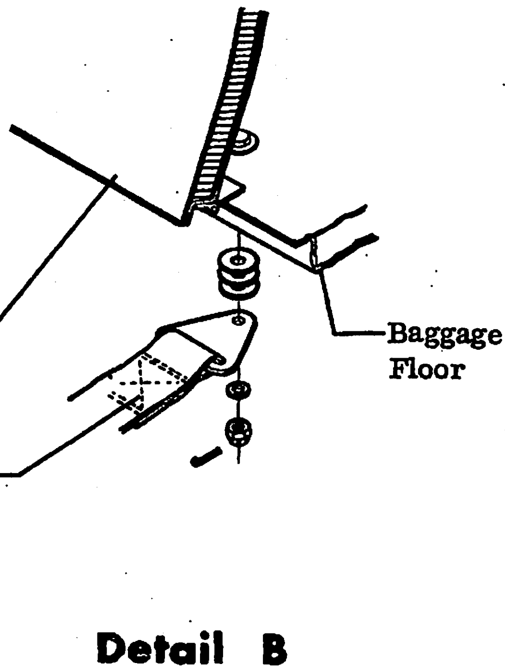


Detail C

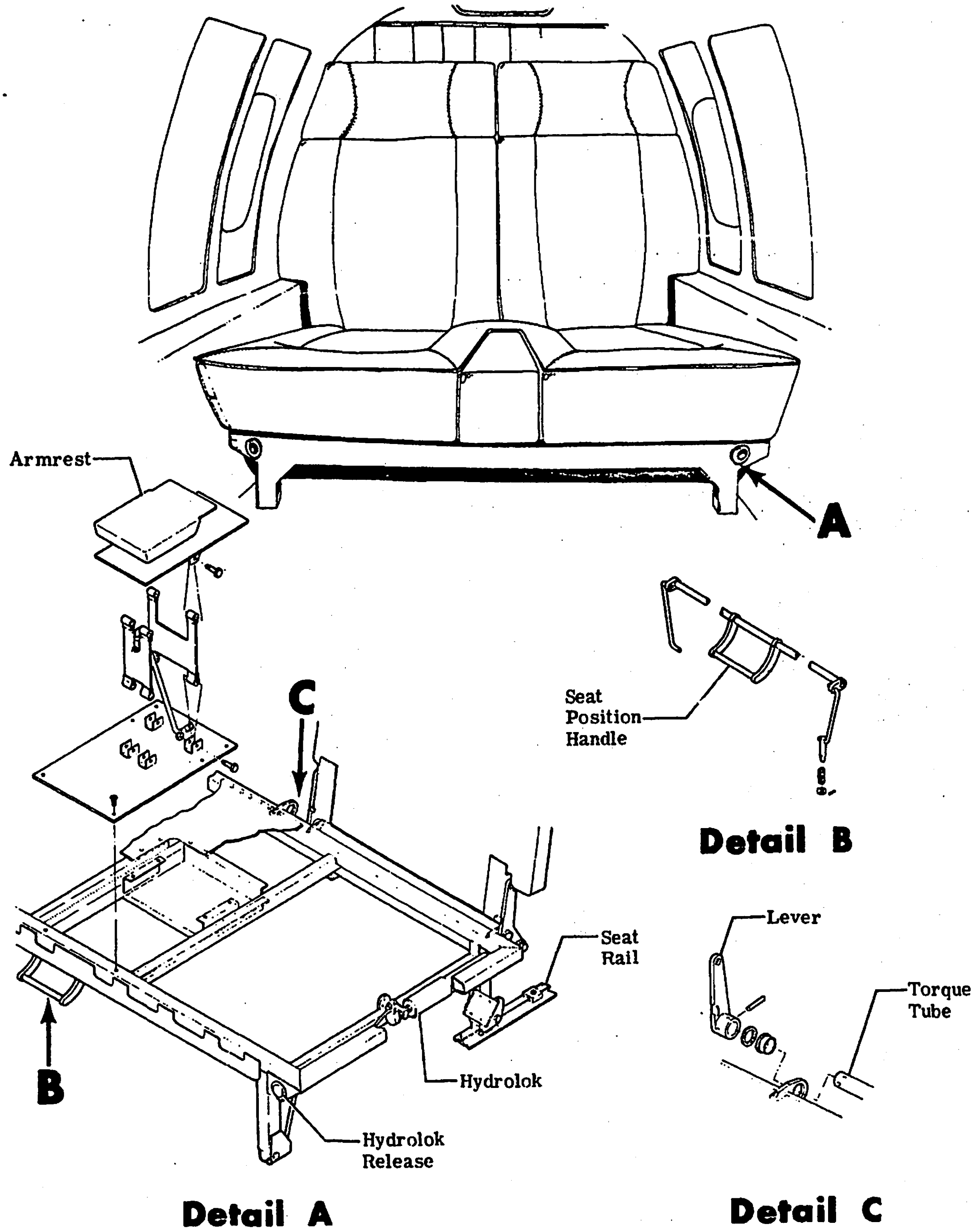
Swivel Seat Assembly
Figure 1 (Sheet 2 of 2)







Divan Seat Assembly
Figure 2



Berthable Divan Seat Assembly
Figure 3

SERVICE CABINET - DESCRIPTION AND OPERATION

1. General
   A. A service cabinet is located in the passenger compartment directly aft of the pilot's seat. Located in the service cabinet are provisions for three one-quart vacuum jugs, a plastic cup dispenser, a bottle cabinet, and two compartments for supply storage.
   B. An emergency hatchet, tool kit and fire extinguisher is stowed inside the LH service cabinet.

EMERGENCY EQUIPMENT - DESCRIPTION AND OPERATION

1. General
   A. Emergency equipment is located in the flight and passenger compartments and includes an emergency tool kit, a hatchet and life vests. In addition, a drag chute is mounted on the inside of the tailcone access door.
   B. The emergency tool kit is stowed in the left hand service cabinet. The ratchet and socket contained in the tool kit are used to open the cabin door in the event of electrical or mechanical failure of the cabin door release actuator mechanism.
   C. An emergency hatchet is also stowed in the left hand service cabinet.
   D. Life vests are stowed in a pocket behind the pilot's and copilot's seats and in a compartment under each passenger seat and in the service cabinet. Inflatable life vests are subject to general deterioration due to aging. Experience has indicated that such equipment may be in need of replacement at the end of 5 years due to porosity of the rubber coated material.

LIFE PRESERVERS - MAINTENANCE PRACTICES

1. General
   A. Life preservers, due to aging and chafing during storage, are subject to deterioration. Life preservers should be inspected on an annual basis if the aircraft is flown over land masses only. Aircraft which frequently fly over water should proportionally decrease the time interval between inspection. In no case should inspection interval exceed 2 years. Experience indicates that preservers may be in need of replacement at the end of 5 years due to the porosity of the rubber coated material.

2. Inspection/Check
   A. Inspect Life Preservers
      (1) Remove the life preservers from aircraft, unfold and lay on a flat surface free of sharp protrusions.
      (2) Inspect life preserver for damage or deterioration. A sour smell, hardness or brittleness indicates perished fabric.
      (3) Inspect all patches for condition and security.
      (4) Inspect all stitching for frayed, rotted or torn stitches.
      (5) Inspect all snap fasteners for correct operation, security, and corrosion.
      (6) Assure that instructions are clearly printed and legible.
      (7) Inspect webbing and metal fasteners for proper operation.
      (8) Assure that all metal parts are free of corrosion and damage and that they function properly.
      (9) Check mouth valves and tubing for leakage, corrosion or damage.
      (10) Remove CO_2 cylinder and inspect discharge mechanism.
      (11) Inspect gaskets, valve core, and pull cord for deterioration.
      (12) Perform a weight check of the CO_2 cylinder.
      (13) The weight of the CO_2 cylinder shall not be less than the weight stamped on the cylinder.

      NOTE: Replacement CO_2 cylinders shall be those with joint Army/Navy Specification MIL-C-00601.

      (14) Install acceptable CO_2 cylinder on life preserver.

      CAUTION: DO NOT USE HIGH PRESSURE AIR TO INFLATE PRESERVER. THIS WILL EXCEED THE DESIGN LIMITS OF THE ORAL INFLATION TUBE AND MAY BURST THE LIFE PRESERVER.

(15) Inflate life preserver to 2 psig and let stand for 12 hours at room temperature (constant).

NOTE: Leak test must be performed with the CO_2 holder assembly properly assembled to the life preserver.

(16) If preserver retains adequate rigidity after 12 hours, it is acceptable.
(17) Deflate preserver.
(18) Mark preserver to indicate inspection date.
(19) Assure that the CO_2 actuating lever is safetied.
(20) Refold the preserver and stow.

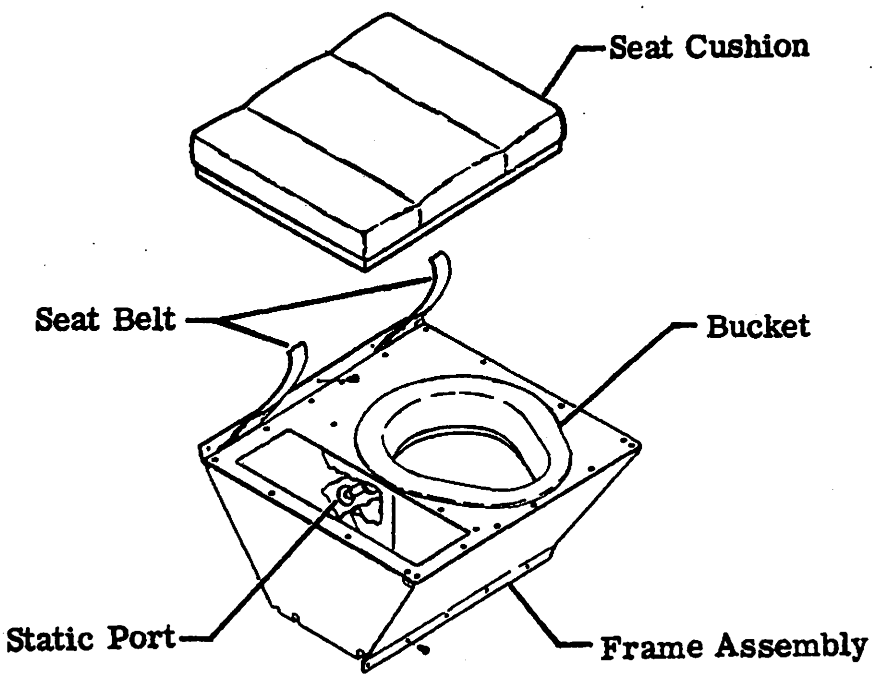
3. Cleaning/Painting

A. Warm soapy water is the only solution to be used in cleaning the life preservers. The preserver should then be thoroughly rinsed, hung up, and allowed to dry. Other cleaners may wash off the printed instructions and deteriorate the fabric.

TOILET - DESCRIPTION AND OPERATION

1. General

A. Toilet facilities are provided under the seat cushion of the forward right passenger seat. The compartment consists of a disposable plastic bag secured to the bottom of the contoured seat with a drawstring. The compartment is under a slight vacuum and is vented overboard. Opaque curtains provide privacy for the cockpit and cabin. The curtains are stowed when unused. For servicing flushing type toilets, refer to SUPPLEMENTARY PUBLICATIONS, in front of this manual.



TOILET SEAT

Toilet Installation
Figure 1

DOWNED AIRCRAFT LOCATOR - DESCRIPTION AND OPERATION

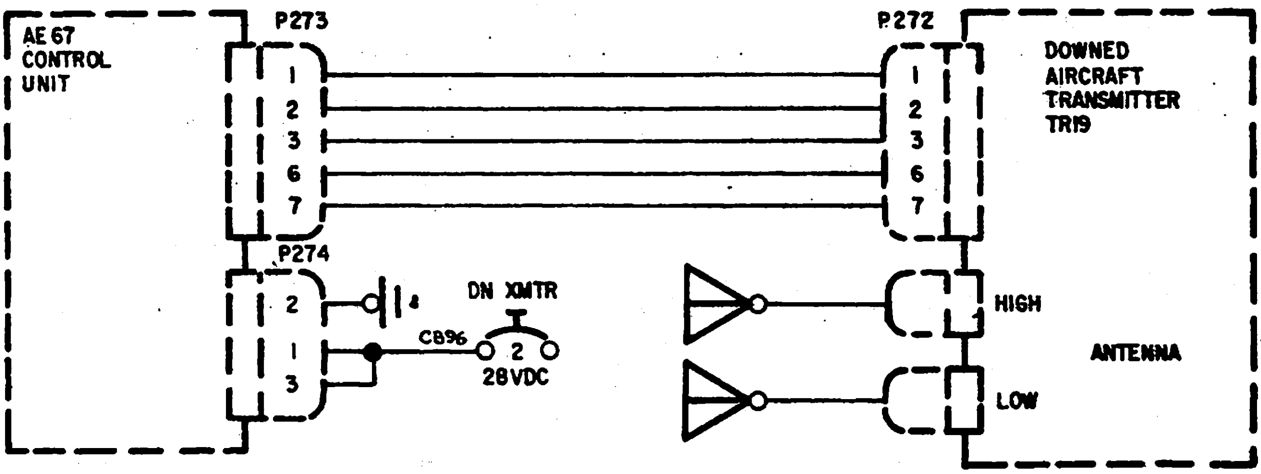
1. General
   A. The downed aircraft locator system consists of a transmitter and dual antennas located in the vertical stabilizer, and a control/charger control head located in the cockpit center pedestal.

2. Description
   A. The transmitter and cover are secured to the LH side of the vertical stabilizer by screws. The transmitter is located forward of spar 2 at approximate W. L. 85.00. The transmitter is a self contained unit with electrical connections for the control/charger head and each antenna.
   B. The antennas are bonded to each side of the vertical stabilizer skin. The antennas are located forward of the transmitter at approximate W. L. 88.00.
   C. The Control/Charger Control unit is located in the cockpit center pedestal. The unit is secured in place by two quick-release fasteners.

3. Operation
   A. The system is powered by 28 vdc thru a 2 ampere circuit breaker. The transmitter batteries are trickle charged thru the control/charger control head. The amber charging light on the control/charger head will be illuminated as long as the transmitter batteries are fully charged. However, the charging light will go out and remain out as long as the batteries are being charged.

      NOTE: At extreme cold temperatures (-30°F or lower) the amber charging light will be illuminated. This is normal and the batteries will maintain a full charge.

   B. The system may be activated in three different ways: (1) by a control switch on the control head; (2) by removing the safety wire and turning the START-AUTO-STOP Switch on the transmitter to START; or (3) by impact.

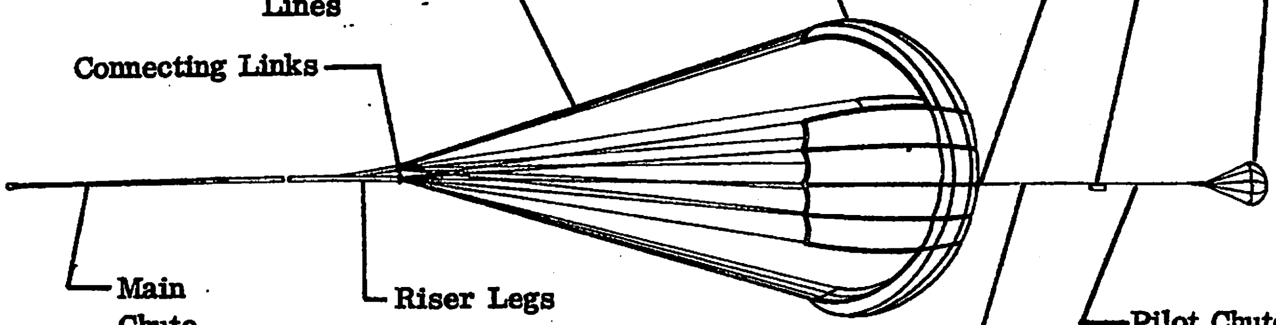


Downed Aircraft Electrical Control Schematic
Figure 1

DRAG CHUTE - DESCRIPTION AND OPERATION

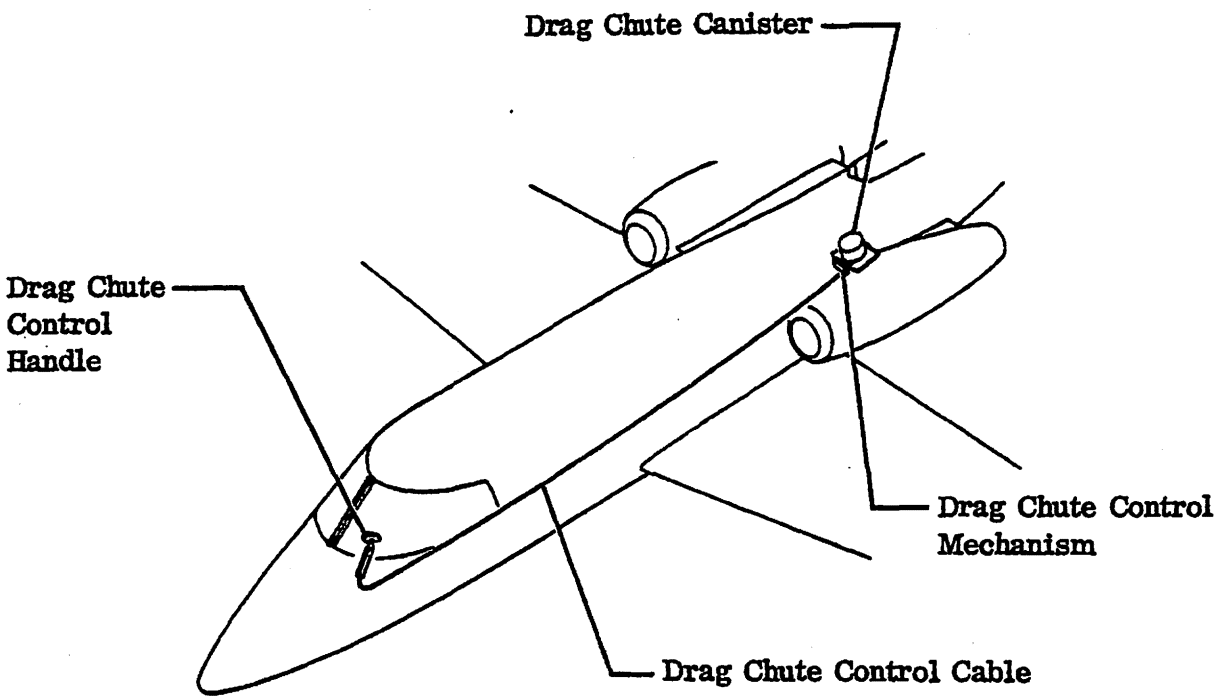
1. Description
   A. The drag chute is optional equipment and will facilitate deceleration in emergency or marginal length runway conditions.
   B. The drag chute control handle is located on the RH side of the pedestal and is equipped with two safeties which must be depressed prior to deployment or jettison. The control cable is routed under the floorboard, under the fuselage fuel tank, and to the drag chute control mechanism.
   C. The drag chute control mechanism is located at the forward edge of the tailcone access door opening.
   D. The drag chute canister contains the packed drag chute assembly, and is installed in the tailcone access door. The canister lid restrains the drag chute in the canister, and is held in place by three pairs of engagement hooks.
   E. The drag chute assembly is contained in the canister and consists of a pilot chute and main chute with their associated risers and suspension lines. The canister lid is connected to the risers between the main chute and the pilot chute.

2. Operation
   A. Both safeties on the drag chute control handle must be simultaneously squeezed into the handle before the handle can be pulled or pushed.
   B. Pulling the control handle to the deploy position causes the control cable to operate the control mechanism in the tailcone. The control mechanism overcenters the chute riser hook, locking it closed, and rotates the unlatching ring, causing the canister lid to rotate slightly and disengage from the canister. The canister lid and pilot chute are then ejected by a spring contained in and attached to the inside of the pilot chute.
   C. Returning the control handle to the "rest/jettison" position causes the unlatching ring to rotate back to its original position. The chute riser hook is moved to center so that it is unlocked but mechanism spring pressure holds it closed. At this point, sufficient tension on the chute riser can overcome the spring pressure, allowing the hook to be pulled open, releasing the drag chute. A manual hook release is incorporated to aid in removal/installation of a packed canister/lid assembly.



Drag Chute Assembly
Figure 1

14-92C-5W



Drag Chute Components Installation
Figure 2

EFFECTIVITY: AIRCRAFT EQUIPPED WITH DRAG CHUTE

DRAG CHUTE - MAINTENANCE PRACTICES

1. Removal/Installation
   A. Remove Drag Chute Canister (See figure 201)
      (1) Lower tailcone access door.

      NOTE: If drag chute is still packed in canister, lower tailcone access door only far enough to gain access to the drag chute attachment hook. This avoids pulling chute riser out of canister inadvertently.

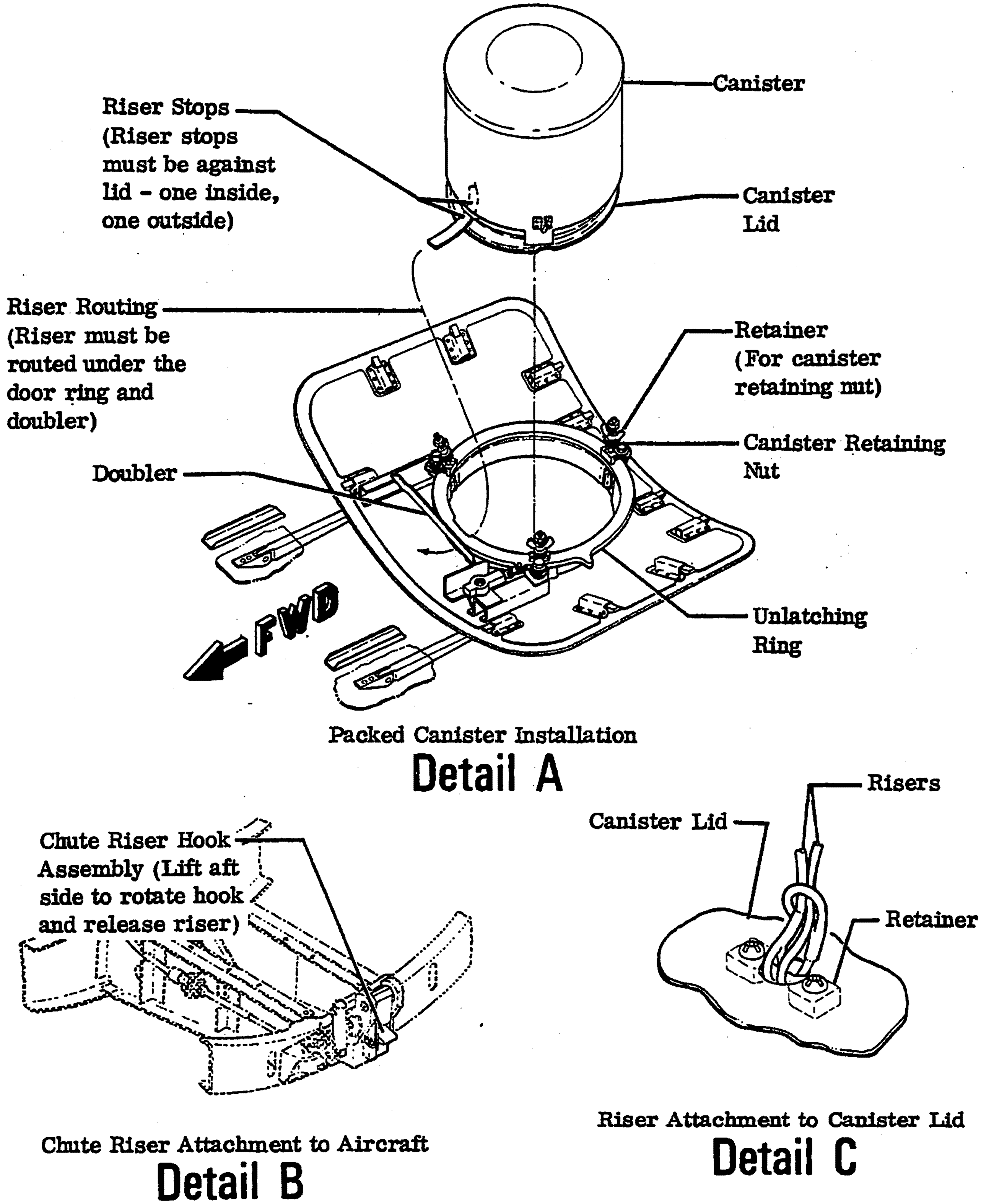
      (2) Raise hook on control mechanism and remove riser from hook. (See detail B.)
      (3) Loosen canister retaining nuts and remove canister and lid assembly from the tailcone access door.
      (4) Rotate lid assembly, taking care to control spring load on lid, and remove lid from canister.
      (5) Remove drag chute from canister.
   B. Install Drag Chute Canister (See figure 201)

   NOTE: Refer to Servicing for drag chute packing procedure.

      (1) Install canister and lid assembly in tailcone access door and secure with canister retaining nuts. Assure that riser protruding from canister is forward.
      (2) Route riser through riser keeper, raise hook of control mechanism (see detail B) and position loop on hook and allow hook to return to rest position.

      CAUTION: ASSURE THAT RISER IS ROUTED UNDER DOOR RING AND DOUBLER, AND THROUGH RISER KEEPER. (REFER TO PLACARD ON TAILCONE ACCESS DOOR FOR CORRECT ROUTING THROUGH KEEPER.) ASSURE THAT ONLY ONE STOP ON RISER IS OUTSIDE CANISTER. (PRESENCE OF TWO STOPS OUTSIDE CANISTER COULD CAUSE DAMAGE TO THE DOOR DURING DRAG CHUTE DEPLOYMENT BY SNAGGING DOOR DOUBLER.)

      (3) Raise and secure tailcone access door.



14-120C
14-121C
14-121C-2

Drag Chute Installation
Figure 201

2. Servicing

A. Pack Drag Chute (See figure 202)

NOTE: Reliable deployment and inflation is critically dependent on closely following this specific folding and packing procedure. The packing panel incorporated in the canister will ease packing and help achieve proper performance.

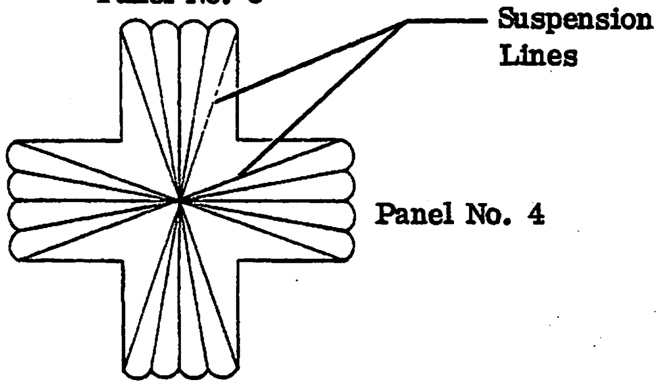
(1) Lay the drag chute on a smooth, clean, dry surface. Ideally the surface should be long enough to completely extend the chute and riser (approximately 55 feet (16.8 meters)). If this is not possible, the riser can be doubled back.

(2) Inspect the drag chute assembly for damage. Refer to Inspection/Check.

NOTE: Each time the drag chute is repacked, a damage inspection should be made of the suspension lines, canopy panels, and risers. A special inspection (refer to Inspection/Check) also is required if deployment was made above 150 KIAS, or if jettison or failure occurred above 100 KIAS.

(3) Separate and straighten the suspension lines of the individual panels to eliminate crossed or twisted lines.

NOTE: For clarity and ease of reference, each of the chute panels must be identified.

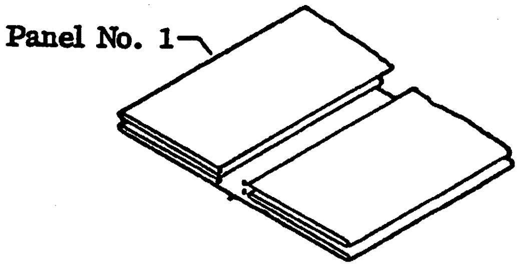


VIEW LOOKING AFT AT OPEN DRAG CHUTE
CANOPY PANEL IDENTIFICATION

Detail A

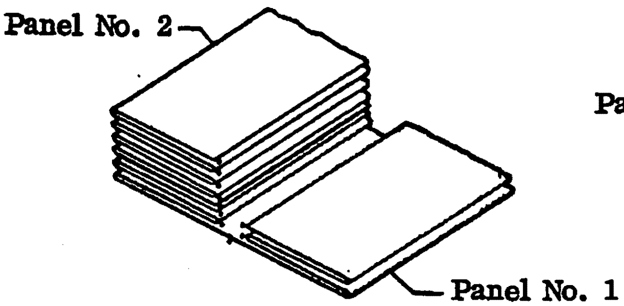
Drag Chute Packing
Figure 202 (Sheet 1 of 5)

NOTE: The (•) shown in details B, C, D, E, and F indicate the points where suspension lines attach to the canopy hem.



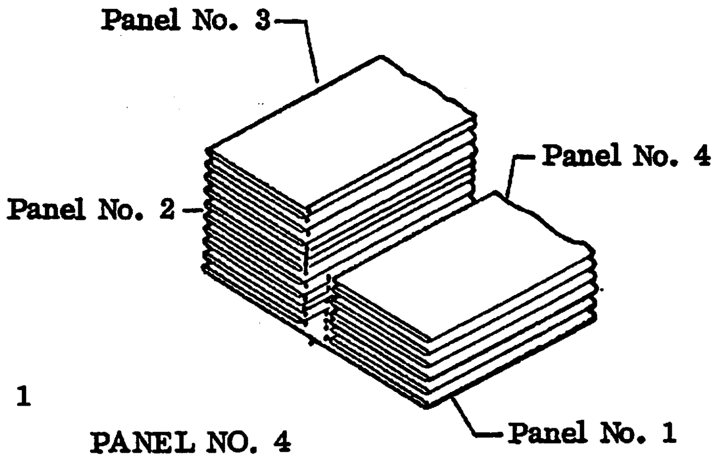
PANEL NO. 1
IN FOLDED CONFIGURATION

Detail B



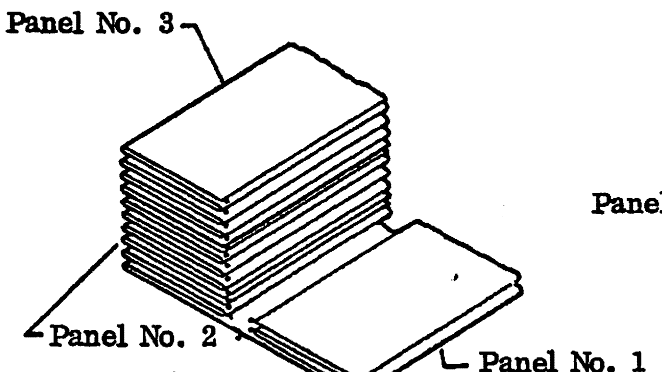
PANEL NO. 2
IN FOLDED CONFIGURATION

Detail C



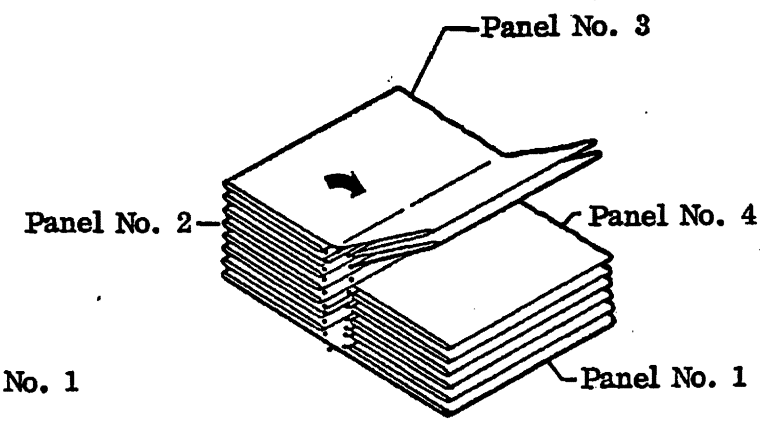
PANEL NO. 4
IN FOLDED CONFIGURATION

Detail E



PANEL NO. 3
IN FOLDED CONFIGURATION

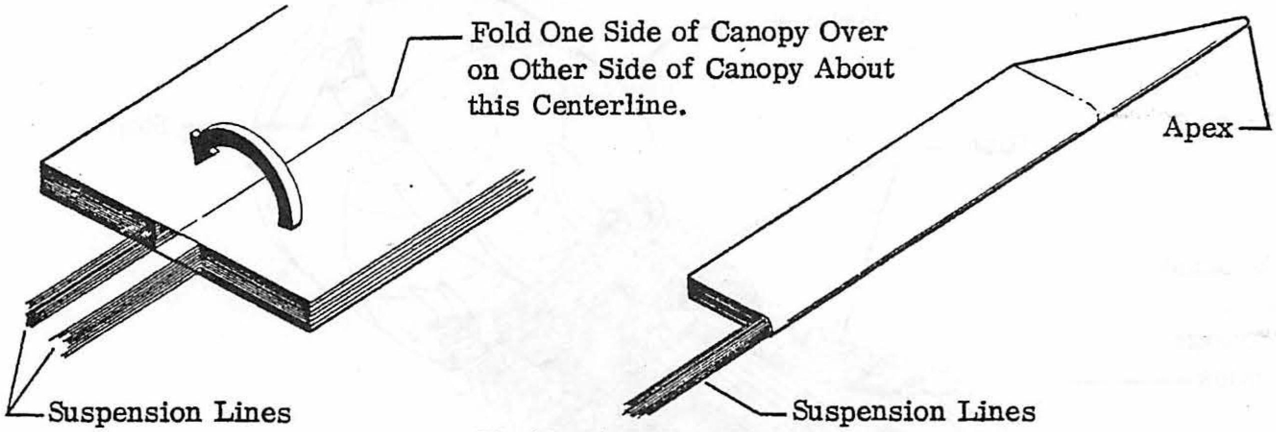
Detail D



PANEL NO. 3 FOLD OVER

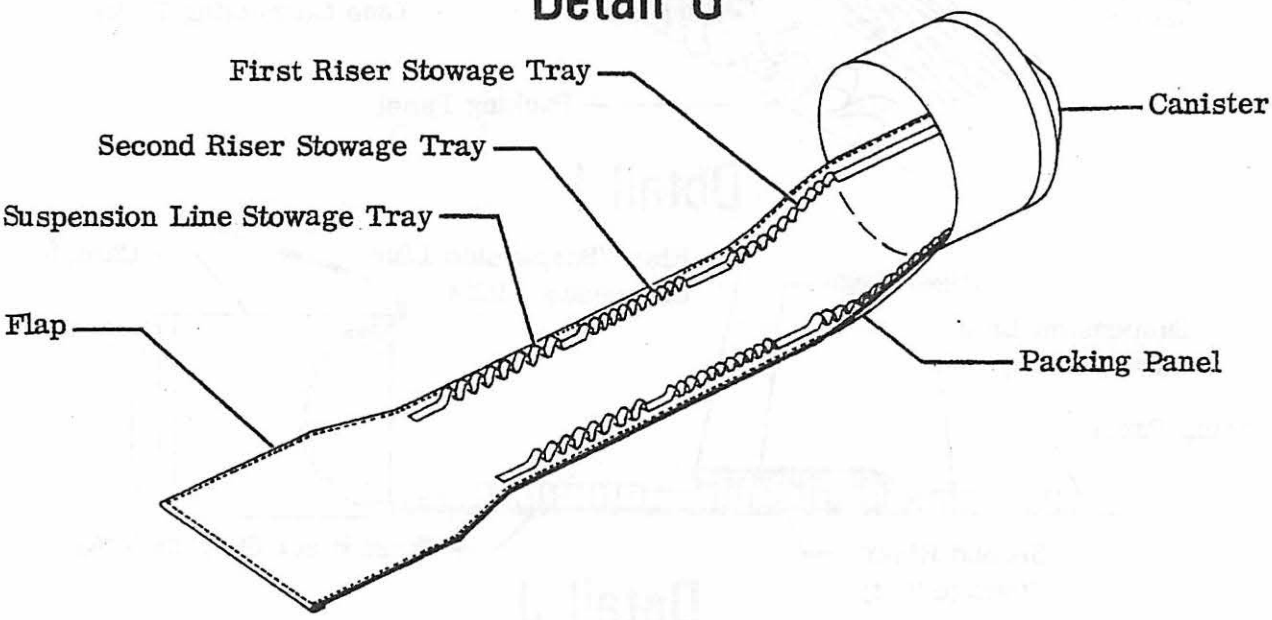
Detail F

Drag Chute Packing
Figure 202 (Sheet 2 of 5)

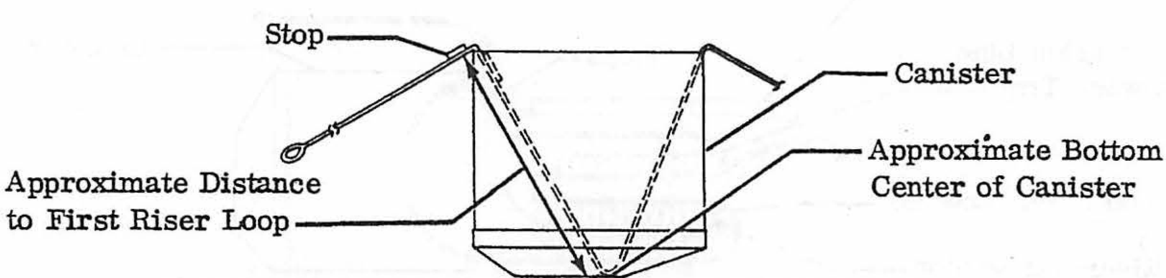


FINAL CANOPY FOLD

Detail G



DRAG CHUTE CANISTER AND STOWAGE PANEL

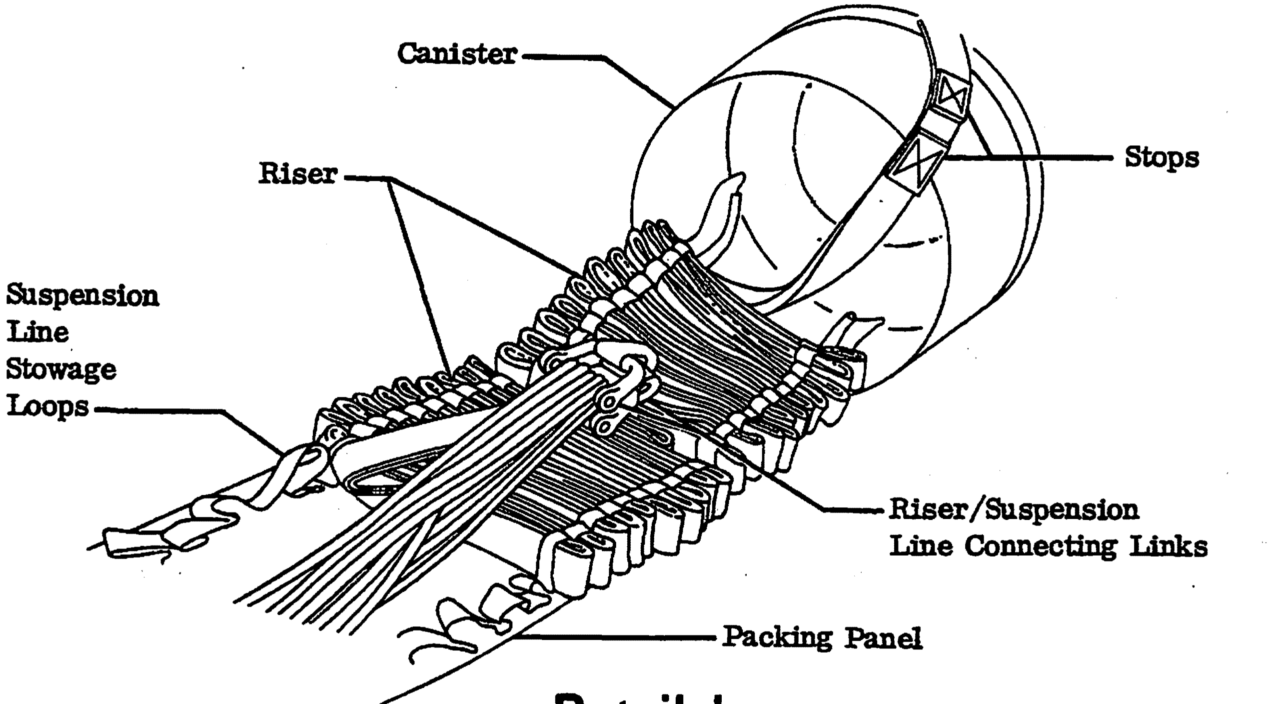


DETERMINING LOCATION FOR FIRST RISER LOOP

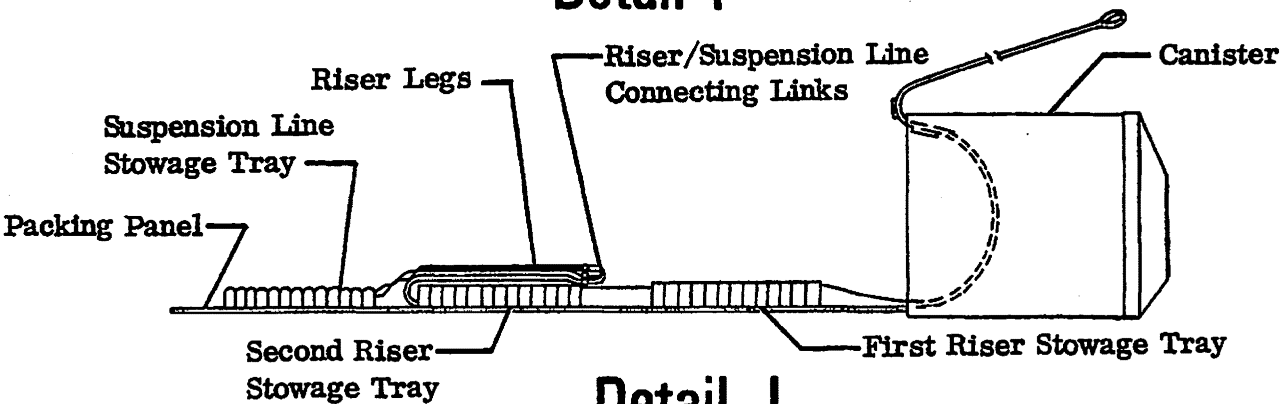
Detail H

Drag Chute Packing
Figure 202 (Sheet 3 of 5)

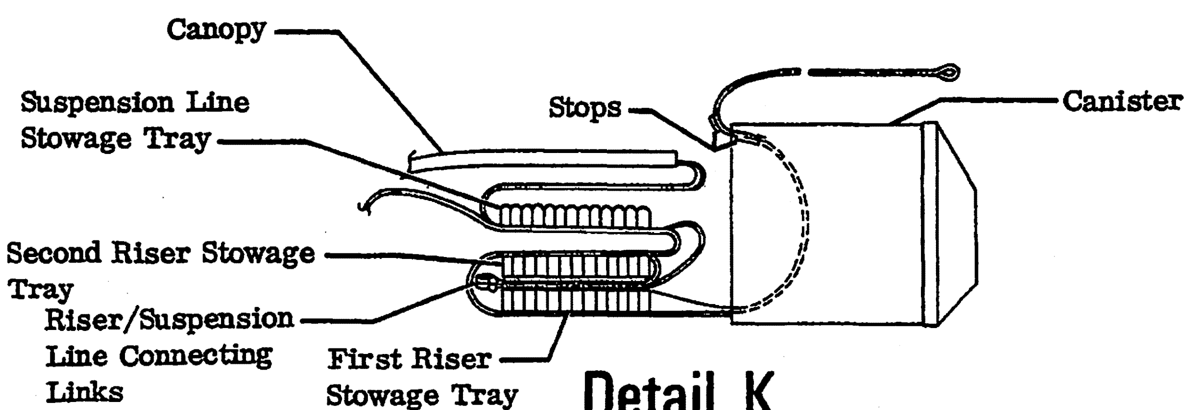
14-121B-1



Detail I



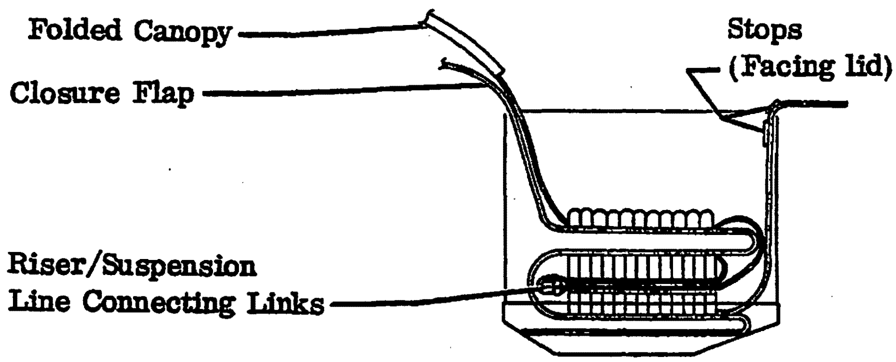
Detail J



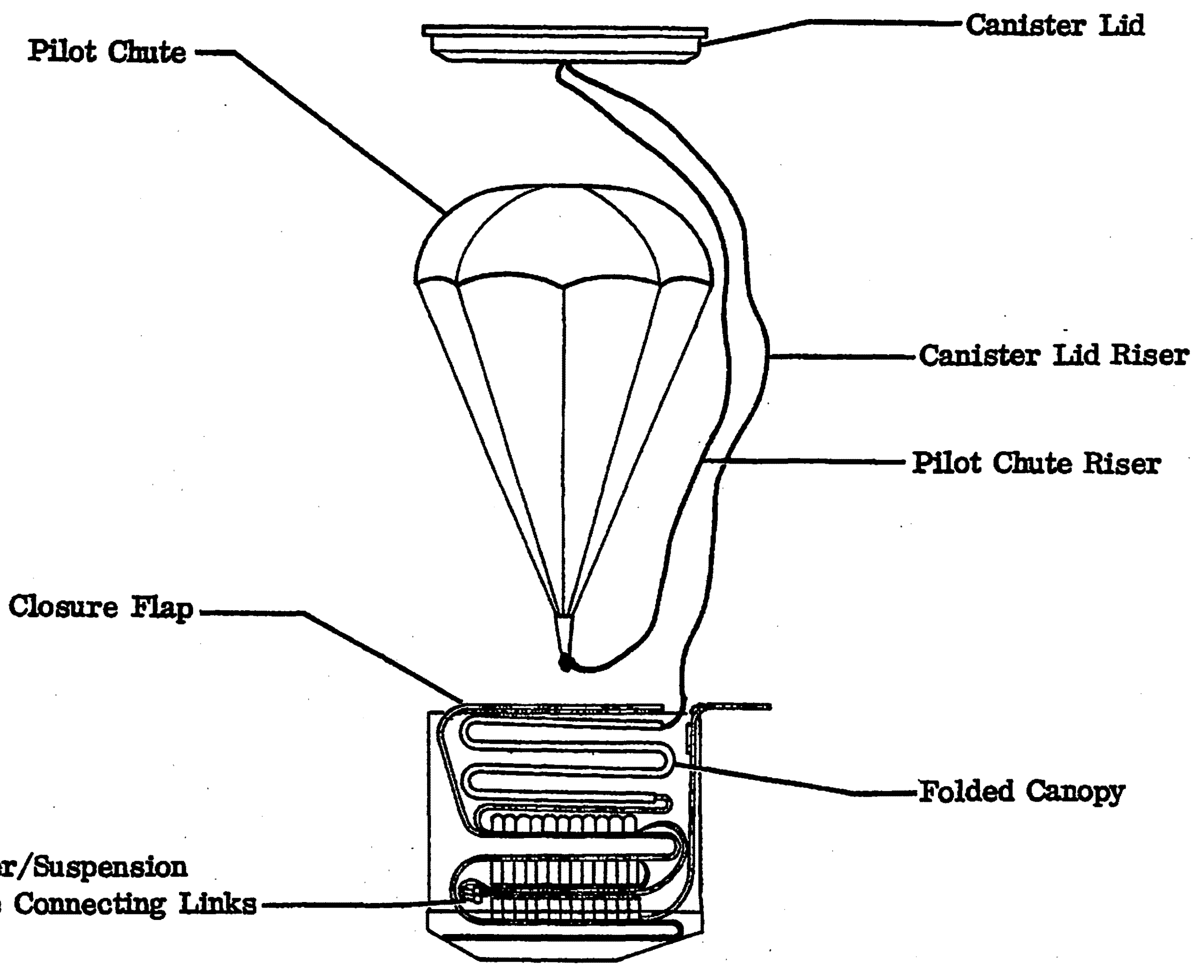
Detail K

14-121B-1

Drag Chute Packing
Figure 202 (Sheet 4 of 5)



Detail L



Detail M

14-121B-1
14-121C-2W

Drag Chute Packing
Figure 202 (Sheet 5 of 5)

(4) Identify one of the panels as number one. The remaining panels should then be numbered in clockwise sequence as shown in detail A.
(5) Separate the suspension lines by panels.
(6) Examine the canopy to assure the correct surface is to the outside. This can be determined by observing the location of the load bearing tapes on the canopy cloth. The canopy should be orientated so that the tapes passing across the canopy are located on the outside.
(7) Assure that the riser is positioned so that all twists are removed.
(8) Assure that riser legs are positioned so that suspension lines from one panel do not cross the suspension lines from any other panel.
(9) Position the number 1 panel with the full panel width laying on the flat folding surface with canopy inside surface up.
(10) Position panel 2 to the left side of panel 1 and panels 3 and 4 to the right side of panel 1.
(11) Fold panel number 1 as shown in detail B. Assure that all folds extend the full length of the panel and on through the apex of the chute.
(12) Fold panel number 2 on top of left half of panel 1 as shown in detail C. The bottom fold of panel 2 and the top fold of panel 1 must be adjacent and panel edges must be aligned.
(13) Fold panel number 3 on top of panel number 2 in the same manner as panel 2 was folded onto panel 1. The numbers 1, 2 and 3 folded panels should appear as shown in detail D.
(14) Fold panel number 4 onto the top right half of panel number 1 as shown in detail E.
(15) Rotate the top two folds of panel 3, 180° clockwise onto top of panel 4 as shown in detail F.
(16) The crown area of the canopy is difficult to fold smoothly and correctly until all panels have been folded as outlined in steps (11) through (15). All folds should now be checked to assure they extend through the crown area to the apex of the chute.
(17) The final step in the canopy folding process is accomplished by folding the canopy on the center line. This is accomplished by rotating those folds on one side of the center line 180°, and placing these folds on top of the other half of the folded canopy. This may be done in either direction, clockwise or counterclockwise. When this folding step is complete, the canopy will be positioned with all suspension line/canopy attachment points on one side of the folded canopy as shown in detail G.
(18) The riser must be oriented with respect to the folded canopy so that no twists exist in the riser. This will necessitate placing the riser legs in the same relative position as the canopy, that is, the leg on the side of the canopy that was folded over should now be on top of the other leg, with no twists in either leg.

(19) The next step is to install the riser in the packing panel stowage trays as shown in detail I. The approximate position from the end of the riser to the first loop is determined by placing the riser over the rim of the packing canister and extending the riser down into the canister. The stop which positions the riser in the canister must be placed to the inside edge of the canister. The point at which the riser makes contact with the center of the canister is the approximate location of the first loop in the stowage panel. This is shown in detail H.

(20) The first loop is put in the tray such that riser orientation is maintained. From this point on, by keeping twists out of the riser, proper orientation will be maintained. The riser loop is forced through the stowage loop on the tray until the riser loop is positioned approximately 1/4 inch (6 mm) past the outside edge of the panel.

(21) The second riser loop is placed in the first stowage loop on the opposite side of the stowage tray. Each riser loop must be positioned as in step 20. Care must be made not to pull preceding riser loops out of the stowage loops when placing the riser in subsequent stowage loops. The process of stowing the riser in the loops of the stowage tray is continued by inserting the risers in each next unfilled stowage loop on alternating sides of the first stowage tray until the space between the first and second stowage trays is reached.

(22) After the eleven pairs of riser loops in the first riser stowage tray are utilized, a space in the stowage loop assembly is reached. This space is to permit folding the stowage tray. The first riser loop in the second stowage tray is placed in the same side as the last riser loop from the preceding series. The length of riser between the stowage trays should be sufficient to facilitate ease of folding. This requires a length of riser sufficient to reach 3/4 of the way across the width of the tray and back to the original side. See detail I.

(23) The remaining length of riser is placed in the second stowage tray in the same manner. Allowance has been provided for packing variations and riser elongation. Therefore, it may not be necessary to use all the stowage loops provided.

(24) Following filling of the second stowage tray, the riser legs must be placed in such a way as to facilitate packing in the canister. This is done by placing the riser legs and the riser/suspension line connecting links as shown in detail J. The links should be approximately centered.

(25) The next step is to fold the suspension lines into the loops of suspension line stowage tray. These loops are loose and serve only to simplify stowage and assure packing and deployment reliability. The first suspension line loop should be selected so that sufficient slack (approximately 2 inches (50.8 mm)) will be left in the lines to allow proper folding of the tray when packing.

(26) Insert the suspension lines into stowage loops. Just as in the risers' stowage, the suspension lines are stowed by inserting the suspension lines in each next unfilled stowage loop on alternating sides of the stowage tray. The stowed suspension lines should extend beyond the edge of the tray no more than 3/4 inch (19 mm). This process is continued until all stowage loops have been filled. When this has been completed, approximately 12 inches (304.8 mm) of suspension lines will remain unstowed.

(27) The second riser stowage tray is folded over the first riser stowage tray as shown in detail K. The suspension line stowage tray becomes the top of the stacked tray. The riser end should be restored to the original orientation as determined at the beginning of the riser stowage procedure in preparation for final packing steps.

(28) The canopy skirt of the folded parachute should be placed on top of the unstowed suspension line.

(29) The stacked trays in the canister, when properly positioned, will appear as shown in detail L. Smooth the flap around the circumference of the canister and fold back over the edge toward the canopy.

(30) The first step in canopy folding is to fold the excess suspension line on top of the stowed lines. The canopy skirt is then placed in the canister at the side where the riser exits and is folded as shown in detail M.

(31) When the canopy is completely folded, the flap is placed over the can opening and pressed down on all sides forming a protective closure. The canister lid riser and the pilot chute riser must be kept outside the flap.

(32) Following the positioning of the closure flap, the pilot chute riser is placed neatly on top of the flap such that the pilot chute spring can be centered on the pack for closure.

(33) Center the lid on the pilot chute spring and while compressing the spring, distribute the pilot chute canopy evenly around the canister. Engage canister lid on canister assuring the drag chute riser exits through the notch provided in the lid. Care should be exercised to assure pilot chute canopy is not jammed into any part of the canister to lid attachment mechanism.

3. Inspection/Check

A. Inspect Drag Chute

(1) Check suspension lines and joints, individually, for cut or frayed cords.

(2) Check canopy panels for deterioration or cuts.

(3) Check main chute riser joints and riser for weak members.

(4) Replace any weak or damaged chute component.

B. Inspect Structure

NOTE: This inspection is required: if the drag chute was deployed above 150 KIAS, or if jettison occurred above 100 KIAS, or if failure occurred.

(1) Inspect frame structure adjacent to where drag chute control mechanism is mounted.

(2) Inspect keel beam forward of the frame on which control mechanism is mounted.

(3) Inspect structure used to attach keel beam to the frame on which control mechanism is mounted.

4. Approved Repairs

A. Repair Drag Chute (See figure 203)

NOTE: Repairs to the drag chute are permissible provided repair procedures are followed. Repairs to the drag chute are to be performed only by an FAA approved repair station.

(1) Holes or damaged canopy areas may be repaired if the hole or damaged area can be patched using a piece of canopy fabric with a maximum area of 100 square inches (64,516 mm). The patch should be cut to the proper size so that it extends beyond the hole or damaged area in all directions by a minimum of one and one-half inches (38 mm). The patch should be sewn on the inside of the canopy using Federal Standard No. 751a stitch type 301 with 8-11 stitches per inch (25.4 mm). The thread used should be V-T-295, Type I or II, Class 1 or 2, Size E. The edges of the cloth around the damaged area and the edges of the patch should be turned toward each other to prevent material fraying. The patch should be secured to the canopy by sewing two rows of stitches around the damaged area and on the patch.

(2) All fabric used to make canopy repairs should be of the same type and quality as that of the canopy material at the point where the repair is being made. In the event damage occurs at an area of transition from one type of fabric to another, i.e. at the location where a side panel joins the center panel, the material used for the patch should be the same as that of the center panel. The four side panels are fabricated using MIL-C-7020, Type III canopy cloth and the center panel is fabricated from nylon fabric, Pattern Number A25895.

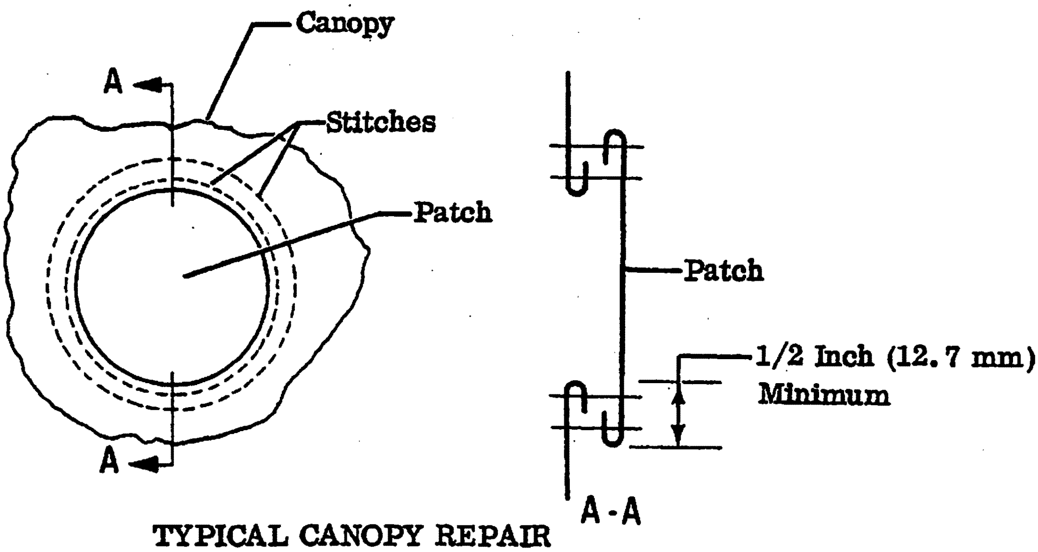
(3) When damage to the canopy occurs at a location where a load tape is located within the damaged area, the repair process is somewhat more difficult. For this type of repair the stitching should be removed from the load tape sufficiently far back from the damaged area to permit the canopy cloth to be turned under. With the stitching removed, the repair patch is sewn as previously described. When the patch has been secured with two rows of stitches sewed around the damaged area, the repair job is completed by sewing the load tapes to the patch. The tapes should be sewed using a Type 301 stitch with 8-11 stitches per inch (25.4 mm). The thread used should be V-T-295, Type I or II, Class 1 or 2, Size E. Two rows of stitching should be used to coincide with the original two rows of stitching on the load tape. The stitching used to secure the load tape to the repaired canopy should be of sufficient length to extend a minimum of two inches (50.8 mm) beyond the point where the stitching had been removed to permit installing the patch. Therefore, at the locations where the original stitching on the load tapes joins the repaired area, double stitching will be provided.

(4) If the damage inflicted to the parachute caused holes in the canopy or damaged areas which cannot be patched as described above with a patch containing an area of 100 square inches (64,516 square mm) the damaged canopy should be replaced rather than repaired.

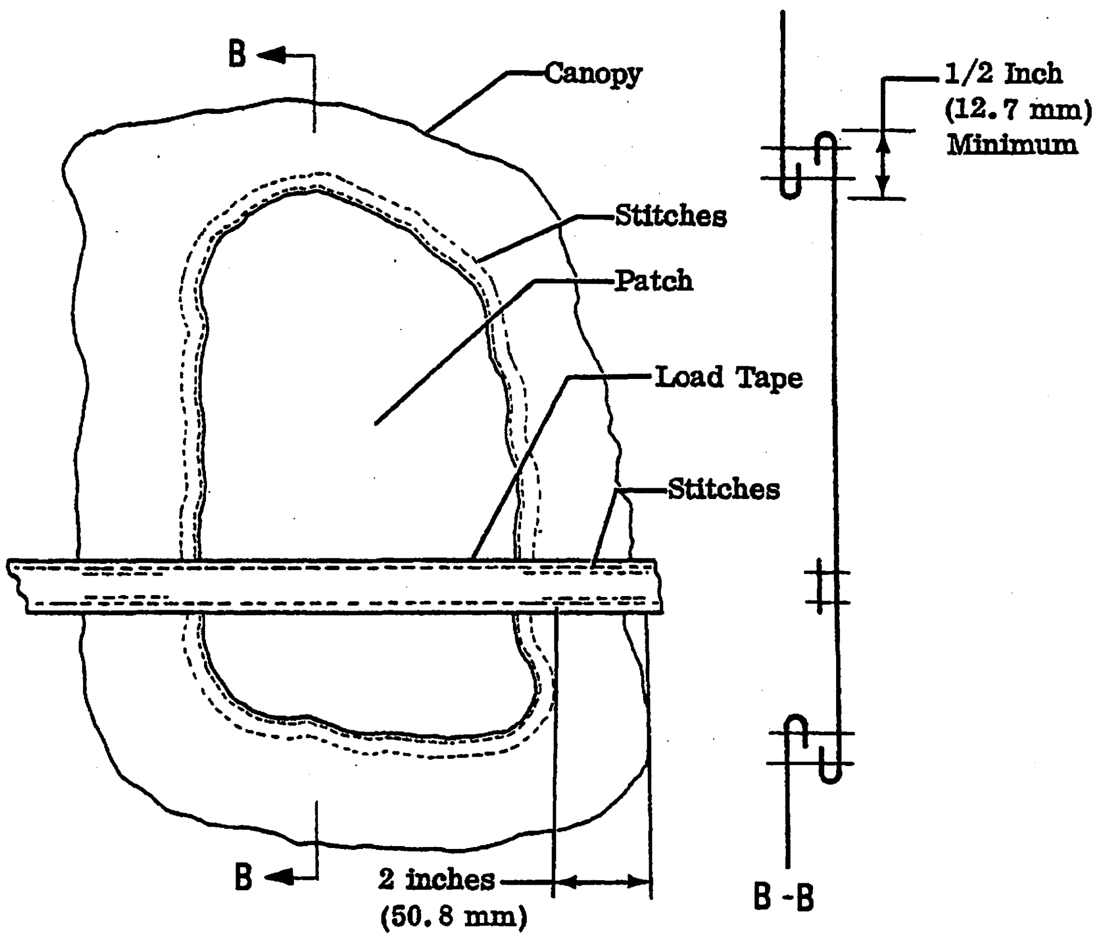
(5) In the event a load tape is damaged, that load tape should be reinforced with a length of tape, MIL-T-5038, Type IV, 5/8 inch (16 mm) wide. The tape should be of sufficient length to extend 6 inches (152 mm) in both directions beyond the damaged location. The reinforcement tape should be sewed on as shown using Type 301 stitch with 8-11 stitches per inch (25.4 mm). Nylon thread, V-T-295, Type I or II, Class 1 or 2, Size E should be used to sew the tape. A two-point W-W stitch pattern should be used to attach the reinforcement tape.

(6) In the event of load tape damage at a location where two load tapes cross and damage has been incurred to both load tapes, the canopy should be replaced rather than repaired.

(7) Frayed or damaged suspension lines or damaged suspension line canopy joints should be repaired by replacing the suspension lines. The damaged suspension lines should be carefully taken off the canopy by cutting the stitches which secure the suspension line to the canopy. Prior to attaching the new suspension line, a reinforcement tape, MIL-T-5038, Type IV, 5/8 inch (16 mm) wide, 12 inches (305 mm) long should be sewed on the canopy immediately above the canopy bottom panel edge. The tape should be sewed on with 2 point W-W stitch pattern using a Type 301 stitch with 8-11 stitches per inch (25.4 mm). Nylon thread, Type I or II, Class 1 or 2, Size E, should be used.

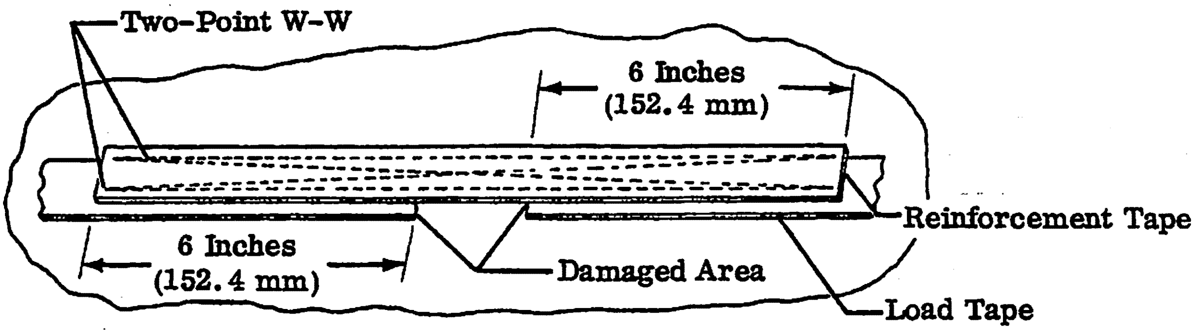


TYPICAL CANOPY REPAIR

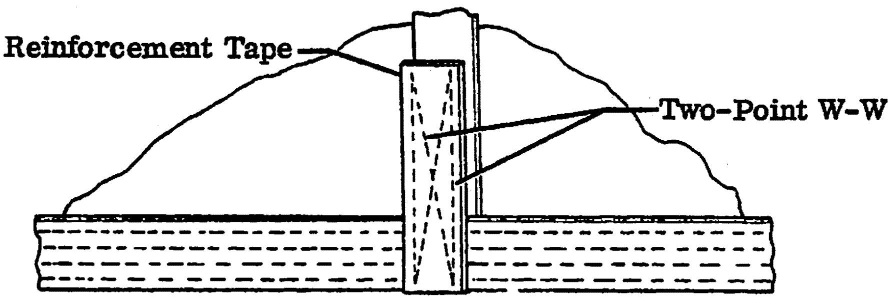


CANOPY REPAIR IN LOAD TAPE AREA

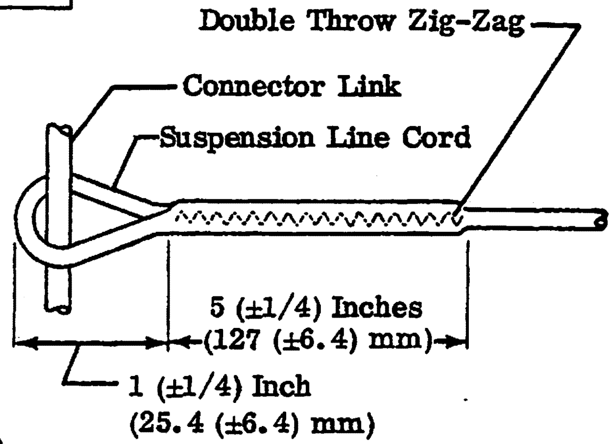
Drag Chute Repair
Figure 203 (Sheet 1 of 3)



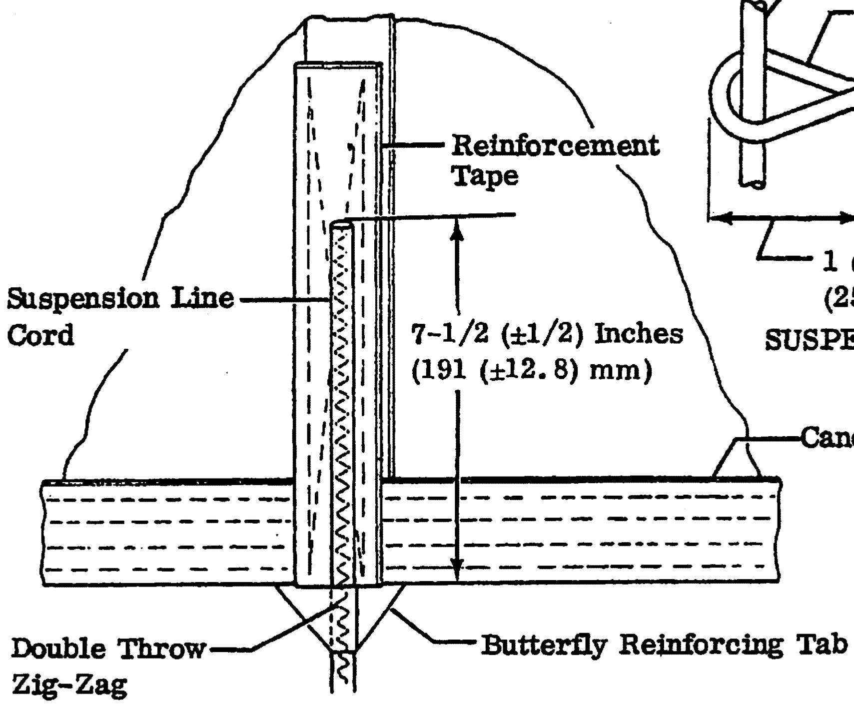
LOAD TAPE REPAIR



SUSPENSION LINE REPLACEMENT

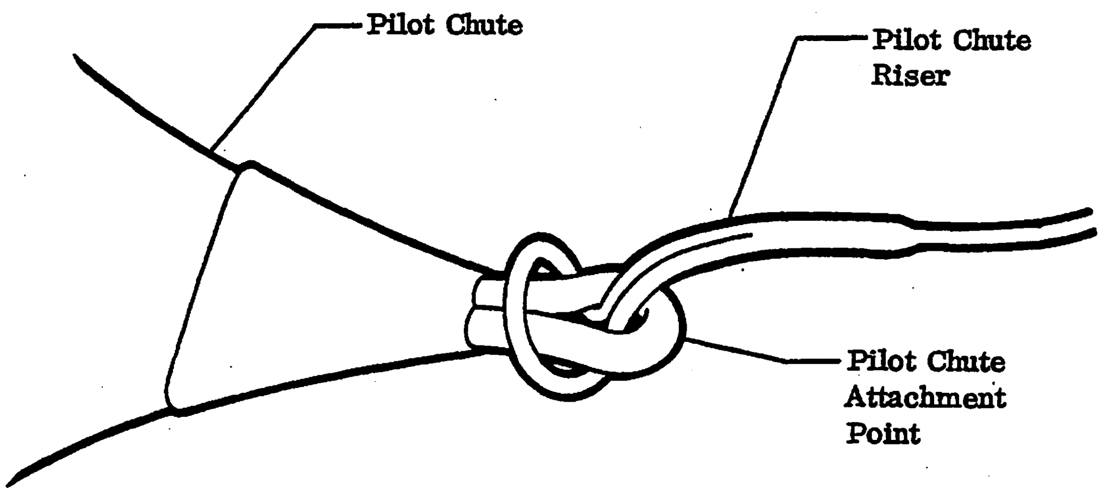


SUSPENSION LINE REPLACEMENT



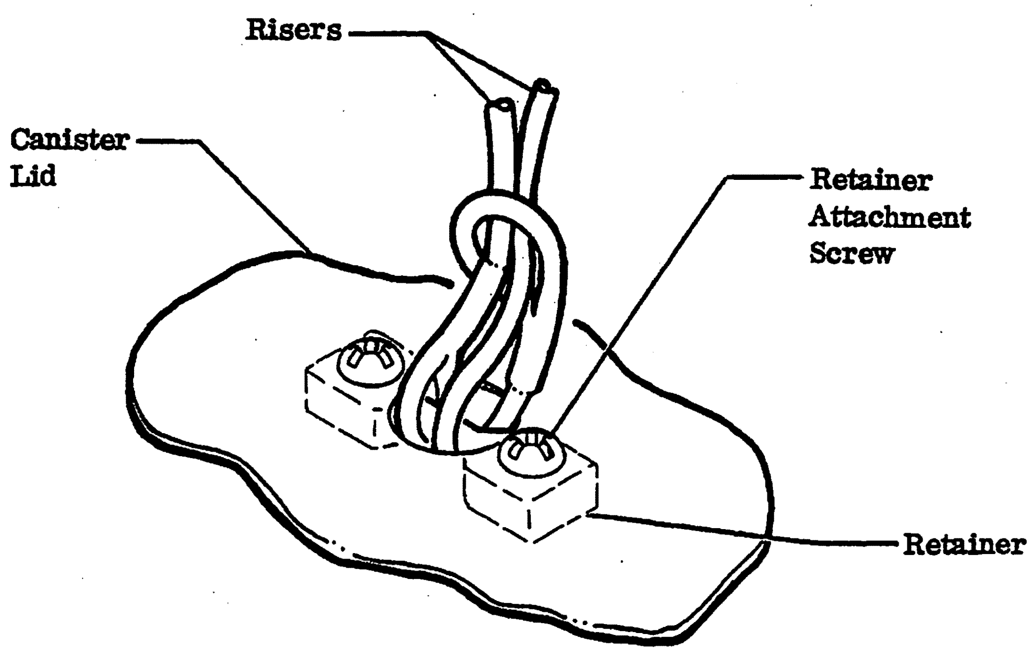
SUSPENSION LINE REPLACEMENT

Drag Chute Repair
Figure 203 (Sheet 2 of 3)



Pilot Chute Riser Attachment to Pilot Chute

Detail A



Risers Attachment to Canister Lid

Detail B

14-121A-1
14-121C-2

Drag Chute Repair
Figure 203 (Sheet 3 of 3)

(8) After the reinforcement tape has been attached, the new suspension line cord can be sewed on. The cord used should be MIL-C-7515, Type III. The cord should be prepared by cutting a length 206 inches (5.23 meters) long, threading one end of the cord around the connector link and back into itself. The cord should then be secured by sewing with a double throw zig-zag stitch with a minimum stitch width of 5/32 inch (4 mm). The double throw zig-zag sewing machine should be set to provide 6-9 edge stitches per inch (25.4 mm). Nylon thread, V-T-295, Type I or II, Class 1, Size F, should be used to sew the suspension line cord. Care must be taken to assure that the new cord occupies the same position on the connector link as was occupied by the replaced cord.

(9) With one end of the suspension line prepared as shown, the other end of the suspension cord should be inserted through the bottom of the butterfly reinforcing tab and placed on the top of the 12-inch (305 mm) reinforcing tape which had previously been sewed to the canopy. The distance from the top of the cord to the bottom of the canopy panel should be 7-1/2 inches (191 mm). When the cord is positioned, the cord must be sewed to the canopy with the same type stitch and thread as was used to prepare the loop in the other end.

(10) If the canopy skirt hem (the edge of the canopy where the suspension lines are attached) is damaged, the canopy should be replaced rather than repaired.

(11) When the risers connecting the main canopy, canister lid, and pilot chute are damaged, the damaged riser should be replaced with a new riser.

(12) Pilot chute riser or canister lid riser attachment is as follows:

(a) Attach the pilot chute riser to the pilot chute by passing one end of the riser through the attachment loop of the pilot chute and then through the loop on the opposite end of the pilot chute riser. Draw the riser tight against the pilot chute attachment loop, and attach the free end to the canister lid. Refer to detail A.

(b) To attach canister lid riser to main canopy, remove bolt from shackle at apex of main canopy, position loop of riser in shackle, and install bolt, passing it through the riser loop.

(c) Attach either or both risers to canister lid by removing retainer attachment screws, and extracting retainer from inside lid. (Pulling on risers while removing screws will aid in controlling retainer during screw removal.) Install risers to retainer as shown in detail B, and insert retainer/riser assembly into canister lid. Align attachment holes in retainer and lid, and install attachment screws. (Pulling risers while installing screws will aid in controlling retainer.)

(13) In the event the main chute riser becomes worn or damaged, the riser should be replaced rather than repaired.

(14) If the pilot parachute is damaged, it should be removed and replaced with a new pilot parachute. The pilot parachute must be attached to the top of the main parachute with the canister lid as described previously.

(15) A damaged stowage tray should be removed and replaced with a new one. The stowage tray can be replaced by removing the two bolts on the inside of the packing container. The new stowage tray should be put into position and secured by inserting the bolts and tightening the nuts.

5. Adjustment/Test

A. Rig Drag Chute Control System (See figure 204)

(1) Lower the tailcone access door.

(2) Position the control handle to the rest/jettison position (control handle fully down).

<u>NOTE</u>: The drag chute control handle is locked unless the two grip safeties are squeezed simultaneously.

(3) Check that the control mechanism crank is full against the stop (rest/jettison position). Loosen telescopic nut and adjust clevis as required.

(4) With the control handle and the control mechanism crank in the rest/jettison position there should be some play in the control handle within its detent.

(5) Squeeze the grip safeties and pull the control handle to the deploy position (control handle fully up).

(6) Check that control mechanism crank is full down against the stop (deploy position). Adjust clevis as required.

(7) With the control handle and the control mechanism crank in the deploy position, there should be some play in the control handle within its detent.

(8) Tighten telescopic nut and install cotter pin.

(9) Squeeze grip safeties and return the control handle to the rest/jettison position.

(10) Raise and secure the tailcone access door.

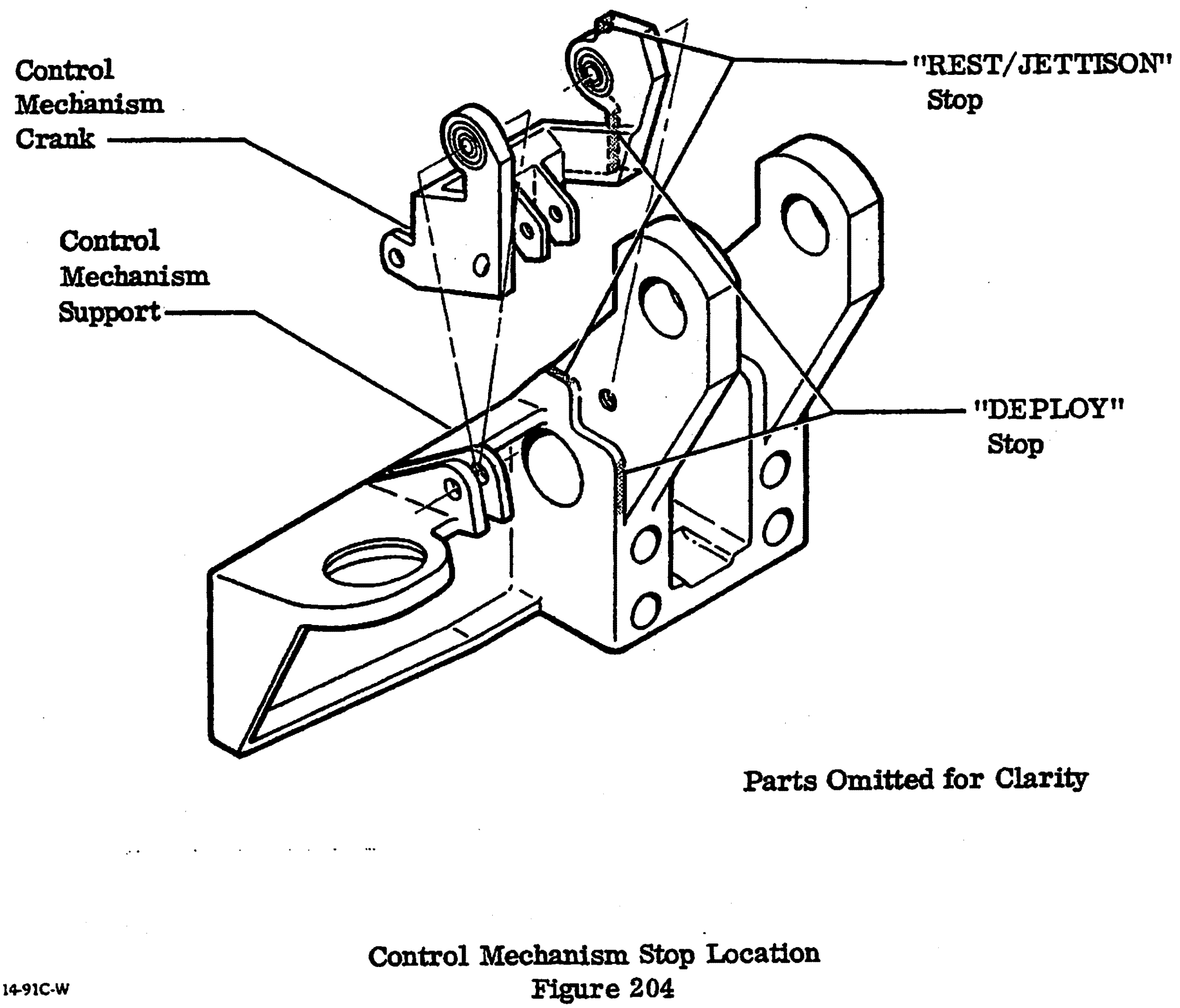
B. Test Drag Chute Control System

(1) Station a man under the tailcone access door to hold and prevent the canister lid from being ejected when drag chute control handle is actuated. The lid assembly must be allowed to pop down only enough to assure disengagement of latching mechanism.

(2) Squeeze grip safeties of control handle and pull up to the deploy position.

(3) Note disengagement of canister lid assembly from canister.
(4) Push lid assembly back up into position.
(5) Squeeze the grip safeties and return the control handle to the rest/ jettison position, thus reengaging the lid latches.

NOTE: In addition to the above functional check, it is recommended that the drag chute be deployed, on landing, at least once during each six month interval. The drag chute should then be inspected and repacked.



Control Mechanism Stop Location
Figure 204